Descriptive Set Theory

MATH 592 · Scribed Course Notes · Part 1: Polish Spaces

Definitions
Results
Examples
Notes

green text marks a term or notation at the point where it is defined

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1 Course Overview

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Note 1. The abbreviations used in the manuscript have been expanded in this edition.

Note 2. Text set in blue (or red) reproduces the scribe's own annotations, written in blue (or red) ink in the manuscript.

Note 3. The marker "Source manuscript page n" indicates where page n of the handwritten manuscript begins.

Note 4. Following the colour convention of the lecturer's notes, a term or a piece of notation is set in green at the point where it is defined, e.g. "a Polish space is ..." or "$[T] := \dots$". Italics are reserved for ordinary emphasis.

Note 5. Results are numbered within sections, and a reference such as Theorem 2.9 is a clickable link to the result it cites; the number is set in this colour so that links are recognizable. Section references like §4.2 are links too.

Relevant readings as prerequisites.

- Analysis 1: § Topology and definitions.
- Analysis 2: § Metric spaces.
- Folland, *Real Analysis*, Chapter 4.

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2 Polish Spaces

Definition 2.1

A Polish space is a topological space which is completely metrizable (i.e. admits a complete compatible metric) and second countable (i.e. the topology admits a countable basis; equivalently, separable). □

Remark. We define it as a topological space and not a metric space, as we don't care about a particular metric, but only about its existence. Also, our maps will be homeomorphisms but not isometries. We may choose a metric that is convenient to us. For instance: □

Proposition 2.2 (truncating the metric)

If d is a complete metric compatible with the topology of a space X, then so is

$$d_{\leq 1}(x, y) := \min\left(d(x, y),\, 1\right).$$

Proof. Exercise. □

Example 2.3

(a) For all $n \in \mathbb{N}$, $n := \{0, 1, \ldots, n-1\}$ is Polish with the discrete topology. More generally, any countable set, like $\mathbb{N}$, equipped with the discrete topology is Polish. By definition, $0 := \varnothing$.

(b) $\mathbb{R}$, $\mathbb{C}$, $\mathbb{R}^n$, $\mathbb{C}^n$, $n \geq 1$.

(c) Any separable Banach space, e.g. $\ell^p(\mathbb{N})$ or $L^p(\mathbb{R}, \lambda)$, $1 \leq p < \infty$ (λ is the Lebesgue measure). In particular, any separable Hilbert space, $\ell^2(\mathbb{N})$ or $L^2(\mathbb{R}, \lambda)$.

(d) Closed subsets of a Polish space are Polish (witnessed by the same complete metric as for the whole space).

(e) $(0, 1)$: indeed $(0, 1) \cong \mathbb{R}$ homeomorphically via the tangent function,

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so the preferred metric is the one transported from $\mathbb{R}$ by the homeomorphism, instead of the metric of $\mathbb{R}$ restricted to $(0, 1)$. □

Proposition 2.4 (Closure properties)

(a) The completion of any separable metric space is Polish.

(b) Countable disjoint unions of Polish spaces are Polish under the disjoint union topology.

(c) Countable products (with the product topology) of Polish spaces are Polish.

Proof. (a) Suppose (X, d) is separable metrizable. Complete (X, d) by the Cauchy sequence construction into $(\widehat{X}, \widehat{d})$ with the complete metric defined as

$$\widehat{d}([x], [y]) := \lim_{n \to \infty} d(x_n, y_n),$$

each of $[x]$, $[y]$ being an equivalence class of Cauchy sequences.

For (b), (c), let $(X_n)_{n \geq 1}$ be a countable sequence of Polish spaces, and d_n a complete compatible metric on X_n with $d_n \leq 1$, $\forall n$. For $X = \bigsqcup_{n \in \mathbb{N}} X_n = \bigcup_{n \in \mathbb{N}} \{n\} \times X_n$, define

$$d((n, x), (m, y)) := \begin{cases} 1, & \text{if } n \neq m, \\ d_n(x, y), & \text{if } n = m. \end{cases}$$

For $X = \prod_{n \in \mathbb{N}} X_n$, define for $x, y \in X$,

$$d(x, y) := \sum_{n \in \mathbb{N}} 2^{-n} \, d_n(x_n, y_n).$$

It is an exercise to show that the d's above are complete compatible metrics for the $\bigsqcup$ and $\prod$ topologies respectively. □

Example 2.5

(a) $\mathbb{R}^{\mathbb{N}}$, $\mathbb{C}^{\mathbb{N}}$ are Polish.
(b) The Hilbert cube $[0,1]^{\mathbb{N}}$ is Polish (and compact by Tychonoff).

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(c) The Cantor space $2^{\mathbb{N}}$ is Polish (and compact), where $2 := \{0,1\}$ with the discrete topology.
(d) The Baire space $\mathbb{N}^{\mathbb{N}}$ is Polish (very noncompact), where $\mathbb{N}$ is with the discrete topology. □

Proposition 2.6 (Binary representation)

Every Polish space X (second countable Hausdorff suffices) admits an injection $b : X \hookrightarrow 2^{\mathbb{N}}$ into the Cantor space such that, for the clopen sets

$$[n \mapsto 1] := \{x \in 2^{\mathbb{N}} : x(n) = 1\},$$

$b^{-1}([n \mapsto 1])$ is open.

(Hausdorff: every two points can be separated by open neighbourhoods.)

Proof. Fix a countable open basis $(U_n)_{n\in\mathbb{N}}$ for X. Define $b : X \to 2^{\mathbb{N}}$ by recording for each x whether it is in U_n or not, i.e. $x \mapsto b(x)$ with

$$b(x)(n) := \begin{cases} 1, & \text{if } x \in U_n, \\ 0, & \text{otherwise.} \end{cases}$$

By definition, $b^{-1}[n \mapsto 1] = U_n$, and injectivity follows from Hausdorffness. □

Corollary 2.7

Polish spaces are at most of cardinality continuum, i.e. $\leq |\mathbb{R}| = |2^{\mathbb{N}}| = \mathfrak{c}$.

Lemma 2.8

If X is a metric space, then closed sets are G_δ (countable intersections of open sets), and open sets are F_σ (countable unions of closed sets).

Proof. For any set $Y \subseteq X$, $\overline{Y} = \bigcap_{n\geq 1} B_{1/n}(Y)$, where

$$B_r(Y) := \{x \in X \mid d(x,Y) < r\} = \bigcup_{y\in Y} B_r(y).$$

Thus if Y is closed, then $Y = \bigcap_{n\geq 1} B_{1/n}(Y)$ and

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the sets $B_{1/n}(Y)$ are open. The open set version follows from taking complements. □

Theorem 2.9

A subset Y of a Polish space X is Polish (in the subspace topology) $\iff$ Y is G_δ. In particular, open and closed subsets of Polish spaces are Polish.

(Recall: $U \subseteq Y$ is open $\iff$ $\exists \widetilde{U} \subseteq X$ open with $U = \widetilde{U} \cap Y$.)

Proof. ($\Leftarrow$) Let's first prove it for an open set $U \subseteq X$. Like $(0,1)$, we will show that U is homeomorphic to a Polish space, specifically a closed subset of $X \times \mathbb{R}$. Take $f : U \to X \times \mathbb{R}$ by

$$x \longmapsto \Big(x,\ \frac{1}{d(x, U^c)}\Big).$$

$f(U)$ is closed: suppose $(x_n, 1/d(x_n, U^c)) \in f(U)$ and $\lim_{n\to\infty}(x_n, 1/d(x_n, U^c)) = (x, r) \in X \times \mathbb{R}$. Because r is finite, $d(x_n, U^c)^{-1} \leq M$ for some $M > 0$, thus $d(x_n, U^c) \geq 1/M \implies d(x, U^c) > 0$ by continuity of the metric $\implies$ $x \in U$, and then $(x, r) = (x, 1/d(x, U^c)) = f(x)$ by the continuity of $d(\,\cdot\,, U^c)$ on U. One has to verify that f is continuous and so is f^{-1} (exercise; we will explain this for the infinite product space next).

Now for a G_δ set $Y := \bigcap_{n\in\mathbb{N}} U_n$, U_n open, likewise consider $f : Y \to X \times \mathbb{R}^{\mathbb{N}}$ by

$$y \longmapsto \Big(y,\ \big(1/d(y, U_n^c)\big)_{n\in\mathbb{N}}\Big).$$

This is again an embedding with a closed image. Notice, from the metric compatible with the countable product topology (Proposition 2.4), that convergence in the product topology $\iff$ convergence in every component.

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Suppose $y_k \to y \in Y$. Then $1/d(y_k, U_n^c) \to 1/d(y, U_n^c)$ for all n, by continuity of $d(\,\cdot\,, U_n^c)$. Convergence in every component $\implies$ $f(y_k) \to f(y) \in X \times \mathbb{R}^{\mathbb{N}}$.

Closedness: Suppose $f(y_k) \to (y, r) \in X \times \mathbb{R}^{\mathbb{N}}$. Then convergence iff convergence in components $\implies$ $y_k \to y$ and $\dfrac{1}{d(y_k, U_n^c)} \to r_n$. Thus $\dfrac{1}{d(y_k, U_n^c)} \leq B_n$ for some $B_n > 0$. Then $d(y_k, U_n^c) \geq B_n^{-1} > 0$. By continuity of $d(\,\cdot\,, U_n^c)$, $d(y_k, U_n^c) \to d(y, U_n^c) \geq B_n^{-1} > 0$. Thus by uniqueness of limits, $r_n = (d(y, U_n^c))^{-1}$. Thus

$$(y, r) = \Big(y,\ \big(d(y, U_n^c)^{-1}\big)_n\Big) = f\big(\lim_k y_k\big) = f(y) \in f(Y).$$

Lastly, f^{-1} is continuous: by definition of the product topology, the projection $\pi_X = f^{-1}$ is continuous.

Now a closed subset of a complete metric space is complete, hence a closed subset of a Polish space is Polish (Example 2.3), admitting a compatible complete metric d_f on $f(Y) \subseteq X \times \mathbb{R}^{\mathbb{N}}$. Define

$$d_Y(x, y) := d_f(f(x), f(y)) \qquad \text{(the pushforward metric)}.$$

Then d_Y is complete and compatible with the subspace topology: denote by τ_Y the topology induced by d_Y, by τ_X the subspace topology of Y in X, and by τ_f the topology induced by d_f, the same as the subspace topology of $f(Y)$ in $X \times \mathbb{R}^{\mathbb{N}}$. We claim $\tau_Y = \tau_X$. τ_Y is generated by the balls $B_\varepsilon^{d_Y}(y)$.

Then $f(B_\varepsilon^{d_Y}(y)) = B_\varepsilon^{d_f}(f(y))$ is τ_f-open, but f is a homeomorphism $(Y, \tau_X) \xrightarrow{f} (f(Y), \tau_f)$, thus $f^{-1}(B_\varepsilon^{d_f}(f(y))) = B_\varepsilon^{d_Y}(y)$ is τ_X-open. Thus $\tau_Y \subseteq \tau_X$. To show $\tau_X \subseteq \tau_Y$, take

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$x \in U \in \tau_X$. Then $f(U)$ is open in τ_f, thus $\exists V$ τ_f-open such that $f(x) \in V \subseteq f(U)$. Then $x \in f^{-1}(V) \subseteq U$ with $f^{-1}(V)$ τ_Y-open, which implies $\tau_X \subseteq \tau_Y$. Together they imply $\tau_X = \tau_Y$, hence d_Y is compatible with the subspace topology of Y. Lastly, d_Y is complete since d_f is. Hence Y with the subspace topology is Polish.

($\Rightarrow$) (Alexandrov) Let $Y \subseteq X$ be a Polish space itself, and let d_X, d_Y[†] be complete compatible metrics for X, Y respectively. For each $n \in \mathbb{N}^+$, define an open set V_n as the union of all open sets $U \subseteq X$ satisfying

(i) $U \cap Y \neq \varnothing$;
(ii) $\operatorname{diam}_{d_X}(U) < \frac{1}{n}$;
(iii) $\operatorname{diam}_{d_Y}(U \cap Y) < \frac{1}{n}$.

Note that (i) and (ii) give $V_n \subseteq B_{1/n}^X(Y) := \{x \in X : d_X(x, Y) < \frac{1}{n}\}$ ("Y puffed by $1/n$"), where $d_X(x, Y) = \inf_{y \in Y} d_X(x, y)$.

We want to show: $Y = \bigcap_{n \in \mathbb{N}^+} V_n$.

By definition, $\forall x \in Y$, $n \in \mathbb{N}^+$, the ball $B_{1/3n}^{d_Y}(x) \subseteq Y$ is open in Y (by definition of the relative/subspace topology, open in the subspace). So $\exists \widetilde{U} \subseteq X$ open such that $B_{1/3n}^{d_Y}(x) = \widetilde{U} \cap Y$. Then $U := \widetilde{U} \cap B_{1/3n}^{d_X}(x)$ satisfies all three conditions (i)–(iii). Thus $\forall x \in Y$, $x \in U \subseteq V_n$ for all $n \in \mathbb{N}^+ \implies Y \subseteq \bigcap_n V_n$.

Let $x \in \bigcap_{n \in \mathbb{N}^+} V_n$. Then there exist open neighbourhoods $\{U_n\}_n$ of x in X satisfying (i)–(iii). Without loss of generality, replace each U_n with $\bigcap_{i \leq n} U_i$ (a finite intersection of open sets);

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then $\{U_n\}_n$ is a decreasing sequence. Choose $x_n \in U_n \cap Y$. Then the sequence is Cauchy with respect to the metric d_Y in Y, because the U_n's are decreasing and by conditions (ii) and (iii). In d_X, $x_n \to x$ since $x \in U_n$ and $\operatorname{diam}_{d_X}(U_n) < \frac{1}{n}$. In d_Y, $\{x_n\}_n$ also has a limit $x' \in Y$ by the completeness of d_Y and Cauchyness. But convergence is a topological property[*], thus $x_n \to x'$ in the subspace topology of Y is the same as $x_n \to x'$ in the topology of X. X is Hausdorff[*], so the limit is unique $\implies x = x' \in Y$ $\implies \bigcap_{n \in \mathbb{N}^+} V_n \subseteq Y$. □

Remark 2.10 ($*$)

Convergence is a topological property: let (X, τ) be a topological space and $\{x_n\} \subseteq X$; then we say $x_n \to x$ as $n \to \infty$ if for every open neighbourhood $U \ni x$, $\exists N \in \mathbb{N}$ such that $\{x_n\}_{n \geq N} \subseteq U$.

Hausdorff = every pair of points is separated by their own neighbourhoods. Then if $x_n \to x$ and $x_n \to x'$ in X, it is easy to see that $x = x'$ if X is Hausdorff. □

Remark 2.11 (†)

$Y \subseteq X$ is Polish with the subspace topology as long as Y has its own complete metric d_Y that is compatible with the subspace topology from X. It need not be complete with respect to the complete compatible metric d_X of X. □

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Example 2.12

(a) The Cantor space $2^{\mathbb{N}}$ is a closed subset of the Baire space $\mathbb{N}^{\mathbb{N}}$ (trivially).
(b) (Homework) The Baire space $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to a G_δ subset of $2^{\mathbb{N}}$. (Hint: think of how to encode a sequence of naturals as a sequence in $\{0,1\}$: encode the numbers by the number of 0's and use "1" as a delimiter between two natural numbers, i.e. the unary representation of integers.) □

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3 Set-Theoretic Trees

Definition 3.1

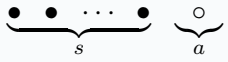
For a nonempty set A, we denote by

$$A^{<\mathbb{N}} := \bigcup_{n\in\mathbb{N}} A^n,$$

where we think of elements of A^n as functions $n \to A$, with $n := \{0, 1, \dots, n-1\}$. Thus we define $A^0 := \{\text{the unique function } \varnothing \to A\} = \{\varnothing\}$. Since functions are also sets of pairs, for $s, t \in A^{<\mathbb{N}}$ it makes sense to write $s \subseteq t$, i.e. the function t extends s, i.e. $|s| \leq |t|$ and $s(i) = t(i)$ for all $i < |s|$, where $|s|$:= the size of the domain of s = the length of s. □

Definition 3.2

For $s \in A^{<\mathbb{N}}$ and $a \in A$, we write sa for the concatenation of s with a, i.e. $|sa| = |s| + 1$ and $(sa)|_{|s|} = s$ (restriction):

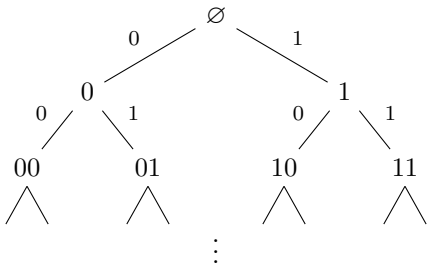


Similarly, for $s, t \in A^{<\mathbb{N}}$, we write st for the concatenation of s with t:



□

Picture. For $A := 2 := \{0, 1\}$, $2^{<\mathbb{N}}$ is the full binary tree:



Definition 3.3

A subset $T \subseteq A^{<\mathbb{N}}$ is called a set-theoretic tree if $T \neq \varnothing$ and T is closed downward with respect to $\subseteq$, i.e. if $t \in T$ and $s \subseteq t$, $s \in A^{<\mathbb{N}}$, then $s \in T$. In particular, $T \neq \varnothing \implies \varnothing \in T$. □

Remark. From a set-theoretic tree T, we obtain a graph-theoretic tree G_T by taking T as the set of vertices and

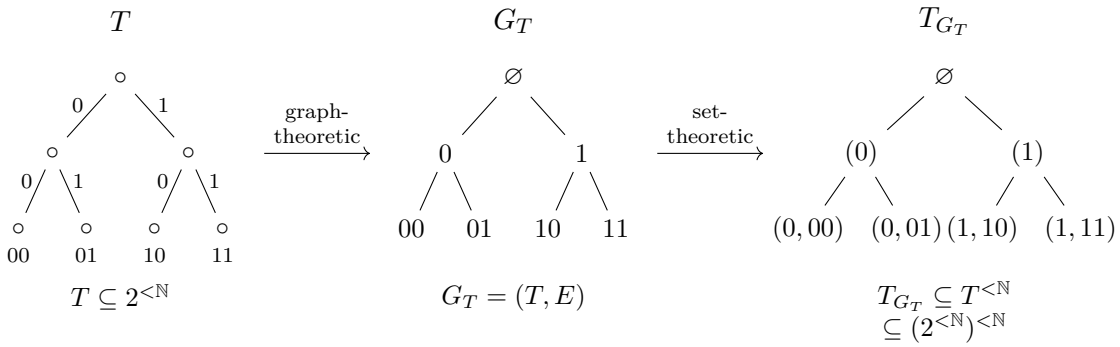
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putting an edge s—t exactly when $t = sa$ or $s = ta$ for some $a \in A$.

Conversely, given a tree $G = (V, E)$ with a root v_0, we obtain a set-theoretic tree T_G on the base set V by

$$s \in T_G \subseteq V^{<\mathbb{N}} \iff s \text{ is a path/walk from } v_0 \text{ to some } v \in V.$$ □

Note. $T_{G_T} \cong T$, via $T \to G_T \to T_{G_T}$, but $T_{G_T} \neq T$. For example, take again $A = 2 = \{0, 1\}$:



More concretely, in T we encode elements by the edges on our walk; in T_{G_T} we encode elements by the vertices $\in T$ on our walk, excluding $v_0 = \varnothing$. □

3.1 Infinite Branches and Closed Subsets of $A^{\mathbb{N}}$

Definition 3.4

For a tree $T \subseteq A^{<\mathbb{N}}$, let $[T]$ denote the set of infinite branches through T, i.e.

$$[T] := \{x \in A^{\mathbb{N}} : x|_n \in T,\ \forall n \in \mathbb{N}\} \subseteq A^{\mathbb{N}}.$$

Conversely, given $Y \subseteq A^{\mathbb{N}}$, we get a tree

$$T_Y := \{x|_n : x \in Y,\ n \in \mathbb{N}\} \subseteq A^{<\mathbb{N}}.$$

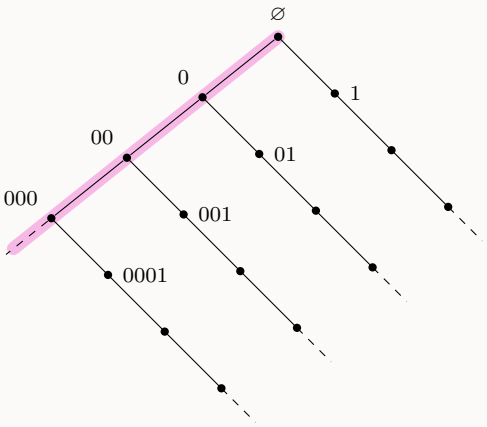
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T_Y is called a pruned tree, in the sense that $\forall s \in T_Y$, $\exists t \in T_Y$ with $s \subsetneq t$. □

Remark. Even though we draw the tree as if the infinite branch is a subtree of it, IT IS NOT. They live in different spaces: infinite branches live in $A^{\mathbb{N}}$, the tree lives in $A^{<\mathbb{N}}$. The truncated parts of the infinite branches belong to the tree $\subseteq A^{<\mathbb{N}}$. □

Example 3.5

$T = \{0^n 1^m : n, m \in \mathbb{N}\} \subseteq 2^{<\mathbb{N}}$.



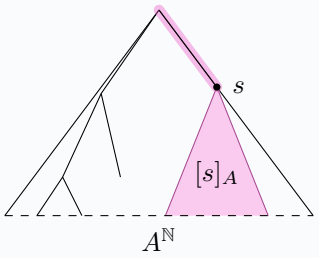
$$[T] = \{0^n 1^\infty : n \in \mathbb{N}\} \cup \{0^\infty\} \subseteq 2^{\mathbb{N}}$$

Conversely, if $Y := \{0^n 1^\infty : n \in \mathbb{N}\}$, then $[T_Y] = Y \cup \{0^\infty\}$. □

Definition 3.6

Give A the discrete topology. In particular, $A^{\mathbb{N}}$ is Polish $\iff$ A is countable. Then the cylinder sets: for $s \in A^{<\mathbb{N}}$,

$$[s]_A := \{x \in A^{\mathbb{N}} : x \supseteq s\} \subseteq A^{\mathbb{N}}.$$



Note that $[s]_A$ is clopen since it is open by definition of the product topology (Definition 4.5), and closed as

$$([s]_A)^c = \bigcup_{\substack{t \in A^{|s|} \\ t \neq s}} [t]_A \subseteq A^{\mathbb{N}}$$

is open. Any set whose membership condition is finite on the number of indices is clopen.

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A complete compatible metric d for $A^{\mathbb{N}}$:

$$d(x,y) := \begin{cases} 2^{-n}, & n \in \mathbb{N} \text{ is the least such that } x(n) \neq y(n), \\ 0, & \text{if } x = y. \end{cases}$$

In this metric, $B_{2^{-n}}(x) = [x|_{n+1}]_A$. Note that the center of this ball can be *any* point in it. More generally, given $s \in A^{<\mathbb{N}}$, $[s]_A$ is the open ball of radius $2^{-|s|+1}$ centered at *any* point in $[s]_A$ (the realtor's metric). □

Proposition 3.7

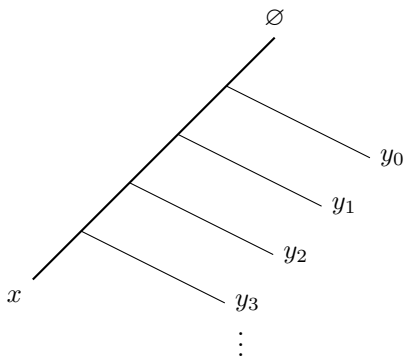
For any tree $T \subseteq A^{<\mathbb{N}}$, $[T]$ is closed. For $Y \subseteq A^{\mathbb{N}}$, $[T_Y] = \overline{Y}$. Thus $Y \subseteq A^{\mathbb{N}}$ is closed $\iff$ $Y = [T_Y]$.

Proof.

$$([T])^c = \{x \in A^{\mathbb{N}} \mid \exists\, n \in \mathbb{N} \text{ such that } x|_n \notin T\} = \bigcup_{n\in\mathbb{N}} \bigcup_{\substack{s\in A^n \\ s\notin T}} [s]_A,$$

which is open. Hence $[T]$ is closed.

Clearly $Y \subseteq [T_Y]$. Take $x \in [T_Y]$. Then $\forall n \in \mathbb{N}$, $x|_n \in T_Y$, i.e. $\exists\, y_n \in Y$ such that $x|_n = y_n|_n$. But the sets $[x|_n]$ form a neighbourhood basis at x (for every neighbourhood $U \ni x$, one can find $x \in [x|_N] \subseteq U$). Thus $y_n \to x$ as $n \to \infty$, thus $x \in \overline{Y}$. So $Y \subseteq [T_Y] \subseteq \overline{Y} \implies [T_Y] = \overline{Y}$, since the closure is the smallest closed set containing Y (we proved this in the setting of metric spaces; it is also true for topological spaces under a more general definition).



□

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Remark 3.8

For countable A, the spaces $A^{\mathbb{N}}$ are perhaps the most important Polish spaces (endowing A with the discrete topology), like $2^{\mathbb{N}}$ or $\mathbb{N}^{\mathbb{N}}$: not only do they have a nice Polish zero-dimensional topology (i.e. one admitting a basis of clopen sets), but they also have combinatorics (discreteness). □

3.2 Trees and Compactness

Question. We already saw that closed subsets of $A^{\mathbb{N}}$ are exactly the sets of infinite branches through trees. But which of them are compact?

Definition 3.9

A tree is called finitely branching if every $s \in T$ has only finitely many immediate extensions, i.e.

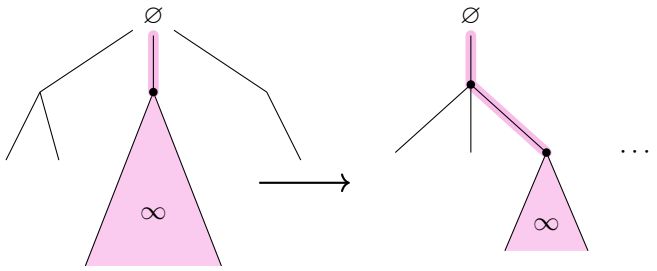
$$\left|\{a \in A : sa \in T\}\right| < \infty,$$

i.e. $\deg(v) < \infty$ for every vertex v. □

Proposition 3.10 (Kőnig's Lemma)

Every finitely branching infinite tree $T \subseteq A^{<\mathbb{N}}$ has an infinite branch, i.e. $[T] \neq \varnothing$.

Proof. Since the tree is infinite, one of the finitely many children is infinite (heavy); reiterate with the subtree through this descendant.



□

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Proposition 3.11 (Homework)

Let $T \subseteq A^{<\mathbb{N}}$ be a pruned tree. Then $[T]$ is compact $\iff$ T is finitely branching. In particular, $Y \subseteq A^{\mathbb{N}}$ is compact $\iff$ T_Y is finitely branching.

[Hint: use a similar approach as for Kőnig's lemma (Proposition 3.10), but think of the finite-covering property and sequential compactness.]

On the other end of the spectrum:

Proposition 3.12

$\mathbb{N}^{\mathbb{N}}$ is not σ-compact (not a countable union of compact sets).

Proof. Let $Y := \bigcup_{n\in\mathbb{N}} K_n$, each $K_n \subseteq \mathbb{N}^{\mathbb{N}}$ compact. We show $Y \neq \mathbb{N}^{\mathbb{N}}$. Let $x_n \in \mathbb{N}^{\mathbb{N}}$ be such that

$$x_n(i) = \max_{y\in K_n} (y(i)),$$

with existence an artifact of the definition of compactness (alternatively, T_{K_n} is finitely branching by Proposition 3.11, so every level has a max). Note: the x_n are not necessarily in K_n. By diagonalization, we get $x \in \mathbb{N}^{\mathbb{N}}$ which dominates each x_n, i.e. $\forall n \in \mathbb{N}$, $\forall^\infty i \in \mathbb{N}$, $x(i) > x_n(i)$ (here $\forall_i^\infty \iff$ for all but finitely many $i \iff \exists m, \forall i \geq m, \dots$). Take

$$x(i) := \max_{n\leq i} x_n(i) + 1 : \quad \begin{array}{llll} x_0 = & x_0(0) & x_0(1) & x_0(2) & \cdots \\ x_1 = & x_1(0) & x_1(1) & x_1(2) & \cdots \\ x_2 = & x_2(0) & x_2(1) & x_2(2) & \cdots \\ \vdots & & & \ddots & \end{array}$$

Note: no Axiom of Choice is used here. AC is needed when we need existence; here the existence is automatic from the explicit construction of x. Hence $x \notin Y$. □

In fact, $\mathbb{N}^{\mathbb{N}}$ is *the* non-σ-compact space, as it is *the* obstruction to σ-compactness:

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Theorem 3.13 (Hurewicz Dichotomy)

For each Polish space X, either X is σ-compact, or $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to a closed subset of X. (Proof left out.)

3.3 Monotone Tree Maps vs. Continuous Functions

We show how to build continuous maps between $[T]$ and $[S]$, for trees T, S, by building monotone maps between the trees T and S.

Definition 3.14

Let S, T be trees on sets A, B, respectively. A map $\varphi : S \to T$ is called monotone if $\forall s_0, s_1 \in S$ with $s_0 \subseteq s_1$, we have $\varphi(s_0) \subseteq \varphi(s_1)$. For such a map φ, let

$$D_\varphi := \left\{x \in [S] : \lim_{n\to\infty} |\varphi(x|_n)| = \infty\right\},$$

i.e. the infinite branches of S whose image by φ in $[T]$ goes to infinity. Then we define $\varphi^* : D_\varphi \to [T]$ by

$$x \longmapsto \bigcup_{n\in\mathbb{N}} \varphi(x|_n) \in [T].$$ □

Proposition 3.15

(a) For each monotone map $\varphi : S \to T$, D_φ is G_δ and $\varphi^* : D_\varphi \to [T]$ is continuous.
(b) Conversely, if $f : D \to [T]$ is continuous for some G_δ subset $D \subseteq [S]$, then $f = \varphi^*$ for some monotone map $\varphi : S \to T$.

Proof. (a): Let $x \in [S]$. Then

$$x \in D_\varphi \iff \underbrace{\forall n}_{\text{countable } \bigcap} \ , \ \underbrace{\exists m}_{\text{countable } \bigcup} \text{ such that } \underbrace{|\varphi(x|_m)| \geq n}_{\text{open}},$$

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i.e.

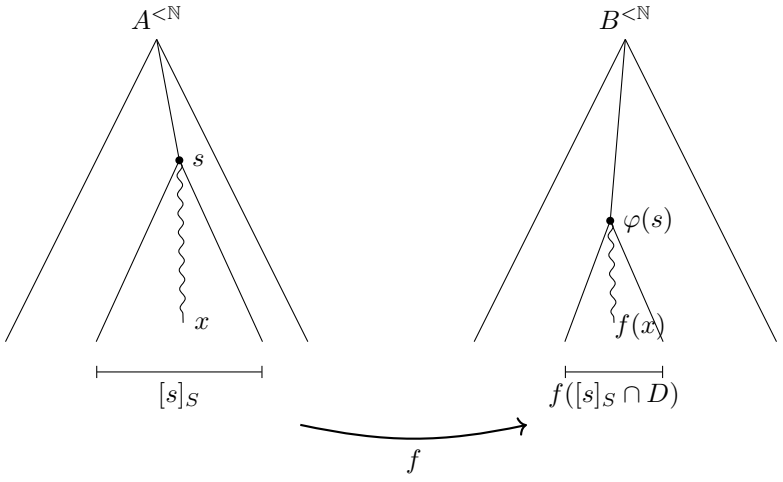
$$D_\varphi = \bigcap_n \bigcup_m \underbrace{\{x \in [S] : |\varphi(x|_m)| \geq n\}}_{U_{nm}},$$

where U_{nm} is clopen as membership of the set is determined by finitely many coordinates ($= m$) (see the note on cylinder sets above); hence D_φ is G_δ. For continuity, observe that if $t \in T$, then for each $x \in D_\varphi$,

$$\begin{aligned} x \in (\varphi^*)^{-1}([t]_T) &\iff \varphi^*(x) \in [t]_T \\ &\iff \underbrace{\varphi^*(x)}_{\text{union}} \supseteq t \\ &\iff \exists n \in \mathbb{N} \text{ such that } \underbrace{\varphi(x|_n) \supseteq t}_{\text{clopen}}, \end{aligned}$$

hence $(\varphi^*)^{-1}([t]_T)$ is open relative to D_φ.

(b) is an exercise; here is the construction. Notation: for $s \in S$, write $[s]_S := \{x \in [S] : s \subseteq x\}$, and for $t \in T$, $[t]_T := \{x \in [T] : t \subseteq x\}$. Idea: $\varphi(s)$ should be the longest possible output in T such that $\forall x \in [s]_S$, $\varphi(s) \subseteq f(x)$, so that $\bigcup_n \varphi(x|_n) = f(x)$. Meanwhile, we want its length to be capped; we will see this is the key to $D_\varphi = D$.



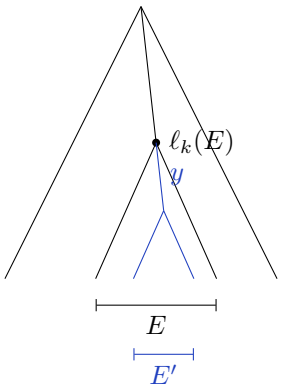
Notion of "longest possible": Let $\varnothing \neq E \subseteq [T]$ and $k \in \mathbb{N}$. Call $t \in T$ a candidate for (E, k) if $|t| \leq k$ and $E \subseteq [t]_T$.

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Then as consequences,

(i) there exists a unique longest candidate, denoted $\ell_k(E)$;
(ii) $\ell_k(E) \subseteq y$ for all $y \in E$;
(iii) if $\varnothing \neq E' \subseteq E$ and $k \leq k'$, then $\ell_k(E) \subseteq \ell_{k'}(E')$;
(iv) if t is a candidate for (E, k), then $|\ell_k(E)| \geq |t|$.

We leave the verifications as exercises to get you used to these definitions.



Cap in the length. Since D is G_δ, write $D = \bigcap_{n\in\mathbb{N}} U_n$, with $U_n \subseteq [S]$ open in $[S]$ and $U_0 \supseteq U_1 \supseteq \cdots$ (if not, take their intersections). Putting $[S]$ in front of the sequence doesn't affect the $\bigcap$, so we assume $U_0 = [S]$. For $s \in S$, define

$$k(s) := \max\{n \leq |s| : [s]_S \subseteq U_n\},$$

the depth to which the cylinder $[s]_S$ is certified to sit inside D. It is well-defined since $n = 0$ qualifies. As a consequence,

(1) $k(s) \leq |s|$.
(2) If $s \subseteq s'$, then $k(s) \leq k(s')$. Check: $[s']_S \subseteq [s]_S \subseteq U_{k(s)}$ and $k(s) \leq |s| \leq |s'|$, then $k(s') \geq k(s)$ by definition.
(3) If $x \in D$, then $k(x|_j) \to \infty$. Indeed, notice that $([x|_j]_S)_{j\in\mathbb{N}}$ forms a neighbourhood basis of x in $[S]$ with the relative topology. Hence $\forall m \in \mathbb{N}$, $\exists N \in \mathbb{N}$ such that $x \in [x|_N]_S \subseteq U_m$. Since $([x|_j]_S)_{j\in\mathbb{N}}$ is decreasing, $\forall j \geq N$, $[x|_j]_S \subseteq U_m$. Hence $\forall j \geq \max(N, m)$,

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$k(x|_j) \geq m$.
(4) If $x \in [S] \setminus D$, then $k(x|_j)$ is bounded. Pick m with $x \notin U_m$. Then $x \notin U_n$ for all $n \geq m$, since the U_n are decreasing. Hence $k(x|_j) < m$, since $x \in [x|_j]_S$ for all j.

Monotone map φ: If $D = \varnothing$, define $\varphi(s) := \varnothing$ for all $s \in S$. Then $D_\varphi = \varnothing = D$ with nothing to prove. Suppose $D \neq \varnothing$.

Along any branch, if $[s']_S \cap D \neq \varnothing$ and $s \subseteq s'$, then $[s]_S \cap D \neq \varnothing$. For every $s \in S$, set

$$\hat{s} := \text{the longest initial segment of } s \text{ such that } [\hat{s}]_S \cap D \neq \varnothing,$$

which exists since $\varnothing$ qualifies. Note that if $[s]_S$ meets D, then $\hat{s} = s$. Define

$$\varphi(s) := \ell_{k(\hat{s})}(f([\hat{s}]_S \cap D)).$$

This is an element of T since, by our choice, $[\hat{s}]_S \cap D \neq \varnothing$; i.e. $\varphi(s)$ is the longest possible $t \in T$ with $|t| \leq k(\hat{s})$ such that f maps all of $[\hat{s}]_S \cap D$ into $[t]_T$.

$\varphi : S \to T$ is monotone: Let $s \subseteq s'$. Then $\hat{s}$ is an initial segment of s' whose cylinder meets D. Thus $\hat{s} \subseteq \hat{s}'$ by maximality of $\hat{s}'$. Hence $[\hat{s}']_S \subseteq [\hat{s}]_S$, giving $f([\hat{s}']_S \cap D) \subseteq f([\hat{s}]_S \cap D)$. On the other hand, $\hat{s} \subseteq \hat{s}' \implies k(\hat{s}) \leq k(\hat{s}')$ by (2). By (iii),

$$\varphi(s) = \ell_{k(\hat{s})}(f([\hat{s}]_S \cap D)) \subseteq \ell_{k(\hat{s}')}(f([\hat{s}']_S \cap D)) = \varphi(s').$$

$D \subseteq D_\varphi$ and $\varphi^* = f$ on D: Recall from Definition 3.14 that $D_\varphi := \{x \in [S] : \lim_{n\to\infty} |\varphi(x|_n)| = \infty\}$ and $\varphi^* : D_\varphi \to [T]$ sends $x \mapsto \bigcup_{n\in\mathbb{N}} \varphi(x|_n)$. Take $x \in D$. Every cylinder set $[x|_j]_S$ contains x,

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hence $\widehat{x|_j} = x|_j$ for all j. Moreover, $\varphi(x|_j) = \ell_{k(x|_j)}(f([x|_j]_S \cap D))$. Since f is continuous, $\forall m\ \exists j_1$ such that $f([x|_j]_S \cap D) \subseteq [f(x)|_m]_T$ for all $j \geq j_1$: indeed, $f^{-1}[f(x)|_m]_T$ is the preimage of an open set in $[T]$, hence open, hence must contain a neighbourhood-basic set $[x|_j]_S \cap D$ at x.

By (3), $\exists j_2$ such that $k(x|_j) \geq m$ for all $j \geq j_2$. For all $j \geq \max(j_1, j_2)$, $f(x)|_m$ is a candidate for $(f([x|_j]_S \cap D), k(x|_j))$ by definition. Thus by (iv), $|\varphi(x|_j)| \geq m$. Hence $|\varphi(x|_j)| \to \infty$, and $x \in D_\varphi$.

Furthermore, $f(x) \in f([x|_j]_S \cap D)$ for all j. Hence by (ii), $\varphi(x|_j) \subseteq f(x)$ for all $j \implies \varphi^*(x) \subseteq f(x)$. But both are infinite branches with domain $\mathbb{N}$, hence they must agree: $\varphi^*(x) = f(x)$ for all $x \in D$.

$D_\varphi \subseteq D$: Let $x \in [S] \setminus D$. Then $|\varphi(x|_j)|$ is bounded by (4), thus $x \notin D_\varphi$. □

Definition 3.16

For a closed subset C of a topological space X, a retraction to C is a continuous function $\pi : X \to C$ such that $\pi|_C = \mathrm{id}_C$ (in particular, π is surjective). If such a retraction to C exists, we call C a retract of X. □

Proposition 3.17

For $A \neq \varnothing$, every nonempty closed subset $C \subseteq A^{\mathbb{N}}$ is a retract of $A^{\mathbb{N}}$. In particular, if $C \subseteq D$ are both closed subsets of $A^{\mathbb{N}}$, then C is a

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retract of D.

Proof. Note that T_C is a pruned tree. Define a monotone map $\varphi : A^{<\mathbb{N}} \to T_C$ such that $\varphi(t) = t$ for all $t \in T_C$ and $D_\varphi = A^{\mathbb{N}}$ as follows, by induction on $|t|$. For $t = \varnothing$, put $\varphi(\varnothing) := \varnothing$. Now take $t = sa$ in $A^{<\mathbb{N}}$, $s \in A^{<\mathbb{N}}$, $a \in A$. Put

$$\varphi(t) := \begin{cases} t, & \text{if } t \in T_C, \\ \varphi(s)b, & \text{otherwise, where } b \in A \text{ is arbitrary such that } \varphi(s)b \in T_C \\ & \text{(it exists since } T_C \text{ is pruned).} \end{cases}$$

By definition, $|\varphi(t)| = |t| \to \infty$ as $|t| \to \infty$, so by definition $D_\varphi = A^{\mathbb{N}}$. Also, $\varphi^*|_C = \varphi^*|_{[T_C]} = \mathrm{id}_C$. Thus φ^* is a retraction of $A^{\mathbb{N}}$ to C. □

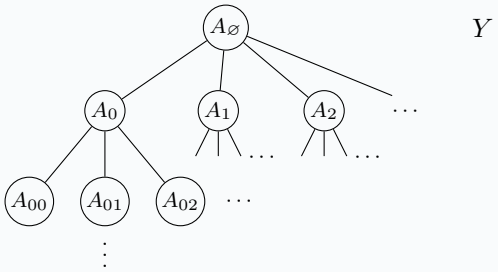
3.4 Luzin and Cantor Schemes

Definition 3.18

A Luzin scheme on a set Y is a sequence $\{A_s\}_{s\in\mathbb{N}^{<\mathbb{N}}}$ of subsets of Y such that

(i) $A_{si} \cap A_{sj} = \varnothing$ for all $s \in \mathbb{N}^{<\mathbb{N}}$, $i \neq j \in \mathbb{N}$;
(ii) $A_{si} \subseteq A_s$ for all $s \in \mathbb{N}^{<\mathbb{N}}$, $i \in \mathbb{N}$

(si, sj being the concatenation of a word with a letter).



If Y is a metric space, then we often require the Luzin scheme $(A_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ to have vanishing diameter, i.e.

$$\forall x \in \mathbb{N}^{\mathbb{N}}, \qquad \lim_{n\to\infty} \operatorname{diam}\left(A_{x|_n}\right) = 0.$$

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In this case, we let

$$D := \left\{x \in \mathbb{N}^{\mathbb{N}} : \bigcap_{n\in\mathbb{N}} A_{x|_n} \neq \varnothing\right\}$$

and define $f : D \to Y$ by $\{f(x)\} = \bigcap_{n\in\mathbb{N}} A_{x|_n}$. This is well-defined, as $\bigcap_{n\in\mathbb{N}} A_{x|_n} \neq \varnothing$ since $x \in D$, and it can contain at most one element since $\operatorname{diam}\left(\bigcap_n A_{x|_n}\right) = 0$. □

Proposition 3.19

Let $(A_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ be a Luzin scheme with vanishing diameter on a metric space Y, and let $f : D \to Y$ be the associated map, where $D \subseteq \mathbb{N}^{\mathbb{N}}$ is as in Definition 3.18.

(a) f is injective and continuous (it need not be a topological embedding, i.e. have a continuous inverse).
(b) If $A_\varnothing = Y$ and $A_s = \bigcup_{i\in\mathbb{N}} A_{si}$, then f is surjective.
(c) If each A_s is open, then f is an embedding.
(d) If Y is complete and $\overline{A_{si}} \subseteq A_s$ for all $s \in \mathbb{N}^{<\mathbb{N}}$, $i \in \mathbb{N}$, then D is closed; in fact, for all $x \in \mathbb{N}^{\mathbb{N}}$, $x \notin D \iff \exists n \in \mathbb{N}$, $A_{x|_n} = \varnothing$. In particular, if every A_s is nonempty, then $D = \mathbb{N}^{\mathbb{N}}$.

Proof. (a) Injectivity by disjointness, (i) and (ii). For continuity, let $x_n \to x$ in D; we show that $f(x_n) \to f(x)$. For each k, $x_n|_k = x|_k$ for all but finitely many n. Hence $f(x_n) \in A_{x|_k}$, $\forall^\infty n$. But then both $f(x), f(x_n) \in A_{x|_k}$, so $d(f(x), f(x_n)) \xrightarrow{n\to\infty} 0$, hence f is continuous.

(b) The first condition says that the first level $= Y$.

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The second condition says that every level, unioned, gives the same thing. Hence $\forall y \in Y$, $n \in \mathbb{N}$, define $x \in \mathbb{N}^\mathbb{N}$ by $x|_n =$ the unique $s \in \mathbb{N}^n$ with $y \in A_s$. Indeed, for $s \supseteq s|_{n-1} \supseteq s|_{n-2} \supseteq \cdots$, we have $A_s \subseteq A_{s|_{n-1}} \subseteq A_{s|_{n-2}} \subseteq \cdots$ by (ii); hence if $y \in A_s$, this ensures $y \in A_{s|_i}$, $i = 1, 2, \ldots, n$. On the other hand, uniqueness follows from (i): at each level, only one such A_s exists, which is in our chain. This gives a unique $s \in \mathbb{N}^n$, and $f(x) = y$.

(c) Enough to observe that for all $s \in \mathbb{N}^{<\mathbb{N}}$,

$$f([s]_\mathbb{N} \cap D) = A_s \cap f(D),$$

which is open relative to $f(D)$. [Topological embedding := homeomorphic onto the image $f(D)$ with the relative topology. Thus seeing that $A_s \cap f(D)$ is open relative to $f(D)$ suffices.]

(d) Fix $x \in \mathbb{N}^\mathbb{N} \setminus D$. The only reason why $x \notin D$, i.e. $\bigcap_n A_{x|_n} = \varnothing$, is that one of the $A_{x|_n}$ was empty: because if $\forall n$, $A_{x|_n} \neq \varnothing$, then

$$\bigcap_{n\in\mathbb{N}} A_{x|_n} = \bigcap_{n\in\mathbb{N}} \overline{A_{x|_n}} \neq \varnothing$$

by the completeness of Y (HW 1). Since $D^c = \bigcup\{[s]_\mathbb{N} : s \in \mathbb{N}^{<\mathbb{N}},\ A_s = \varnothing\}$ is a union of clopen cylinders, it follows that D^c is open, so D is closed. □

Definition 3.20

When the scheme is indexed by $2^{<\mathbb{N}}$ instead of $\mathbb{N}^{<\mathbb{N}}$, it is called a Cantor scheme, and everything above (Definition 3.18, Proposition 3.19) equally holds for such schemes $(A_s)_{s\in 2^{<\mathbb{N}}}$. □

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4 Compact Metrizable Spaces

Compact metrizable space = compact Polish space.

Recall. Compact := the finite covering property. Taking complements and the contrapositive, we get the equivalent formulation:

Definition 4.1

A topological space X is compact if every family $\mathcal{F}$ of closed subsets of X with the finite intersection property (i.e. for every finite $\mathcal{F}_0 \subseteq \mathcal{F}$, $\bigcap_{F\in\mathcal{F}_0} F \neq \varnothing$) has a nonempty intersection: $\bigcap_{F\in\mathcal{F}} F \neq \varnothing$. □

Theorem 4.2 (Heine–Borel)

For a metric space (X, d), the following are equivalent:

(a) X is compact.
(b) X is sequentially compact, i.e. every sequence has a convergent subsequence, converging *in* X.
(c) (X, d) is complete and totally bounded, i.e. $\forall \varepsilon > 0$ there is a finite partition of X into sets of diameter $\leq \varepsilon$; equivalently, there is a finite set $F \subseteq X$ which is ε-dense, i.e.

$$X = B_\varepsilon(F) = \bigcup_{x \in F} B_\varepsilon(x).$$

In particular, every compatible metric on a compact topological space X is complete.

Proof. We give the intuitive picture, which one can formalize into a proof. A detailed proof is in HW 1.

(a) $\Rightarrow$ (b) Suppose not. No converging subsequence $\implies$ every point has an open ball that only contains finitely

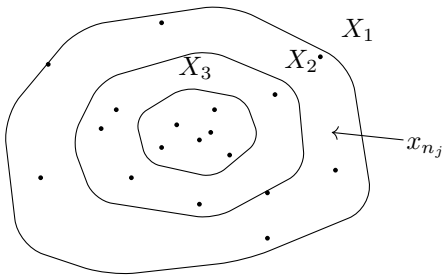
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many elements of the sequence $\implies$ contradicting the finite cover property.

(b) $\Rightarrow$ (a) First show that $\exists r > 0$ such that $\forall x \in X$, $B(x, r) \subseteq U_i$ for some i in the open cover (also proven by contradiction). Use this to construct a sequence of balls $B(x_i, r)$ that are mutually disjoint; then X must be covered by finitely many of them, and thus by the U_i's they are contained in: if X cannot be covered, then the sequence $\{x_i\}$ has no convergent subsequence, since all terms are at a distance $\geq r$.

(a) & (b) $\Rightarrow$ (c) Obviously, (b) $\implies$ completeness, since every Cauchy sequence converges to the limit of its subsequence. (a) $\implies$ total boundedness, observing that $\forall \varepsilon > 0$, $\bigcup_{x \in X} B(x, \varepsilon)$ is a cover, then applying the finite cover property.

(c) $\Rightarrow$ (b) Suppose total boundedness. Pick a sequence $\{x_n\}_n$. By total boundedness and the pigeonhole principle, we can construct a nested descending sequence $\{X_k\}_{k=1}^\infty$ of subsets of X such that $\operatorname{diam}(X_k) \leq \frac{1}{k}$, and a subsequence $\{x_{n_j}\}_j$ of $\{x_n\}$ such that $x_{n_j} \in X_k$, $\forall^\infty j \in \mathbb{N}$, $\forall k \in \mathbb{N}$; i.e. X_k contains infinitely many terms of this subsequence; pass to this subsequence and construct its subsequence in the same way. Pick the diagonal subsequence; this subsequence is Cauchy, hence convergent by completeness.



□

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Corollary 4.3

Compact metrizable spaces = compact Polish spaces.

Proof. ($\Rightarrow$): X is compact metrizable $\implies$ $\exists$ a countable dense subset $\implies$ compact Polish. ($\Leftarrow$) follows from the definition. □

4.1 Facts about Compact (Metrizable) Spaces

Definition 4.4

Let $\{X_i : i \in I\}$ be a family of topological spaces indexed by I. Let

$$X = \bigsqcup_i X_i := \bigcup_{i \in I} \{i\} \times X_i$$

(so that if $X_i = X_j$, still $\{i\} \times X_i \neq \{j\} \times X_j$), with the topology generated by the multiset $\{\{i\} \times U_i \mid i \in I,\ U_i \subseteq X_i \text{ open}\}$, called the disjoint union topology. Equivalently, it is the finest topology on X such that the canonical injections $\varphi_i : X_i \hookrightarrow X$, $\varphi_i(x) := (i, x)$, are continuous. Explicitly, a subset is open $\iff$ its trace is open in $\{i\} \times X_i$ for every i. □

Definition 4.5 (Product topology)

If $\{X_\alpha\}_{\alpha \in A}$ is a family of topological spaces, the product topology on $X = \prod_{\alpha \in A} X_\alpha$ is the coarsest/weakest topology such that the projection maps

$$\pi_\alpha : X \to X_\alpha, \qquad f \mapsto f(\alpha), \qquad \forall \alpha \in A,$$

are continuous. Equivalently, it is the coarsest topology generated by the multiset $\{\pi_\alpha^{-1}(U_\alpha) : U_\alpha \subseteq X_\alpha \text{ open}\}$. Explicitly, a basic open set in $\prod_\alpha X_\alpha$ is one in which all but finitely many coordinates are the whole space X_α, while the remaining finitely many are open sets in their own X_α. □

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Example 4.6

One can verify that the cylinder sets $\{[s]_A : s \in A^{<\mathbb{N}}\}$ give a basis of $A^{\mathbb{N}}$. □

Theorem 4.7

(a) Closed subsets of compact topological spaces are compact (in the relative/subspace topology).
(b) Compact (in the relative/subspace topology) subsets of Hausdorff spaces are closed.
(c) A finite union of compact sets in a topological space is compact. Finite sets are compact.
(d) The continuous image of a compact space is compact. In particular, if $f : X \to Y$ is continuous, X is compact and Y is Hausdorff, then f is a closed / F_σ mapping, i.e. it maps closed (respectively F_σ) sets to closed (respectively F_σ) sets. But note that open map $\neq$ closed map.
(e) A continuous injection from a compact space into a Hausdorff space is an embedding (i.e. a homeomorphism with the image).
(f) A disjoint union of finitely many compact spaces is compact with respect to the disjoint union topology.
(g) (Tychonoff's Theorem) Any arbitrary product of compact spaces is compact.

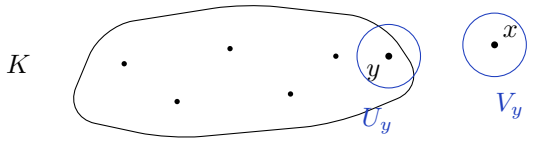
Proof. (a), (c)–(f) are trivial from the definitions once we verify (b): Let $K \subseteq X$ be a compact subset (with the relative topology) of a Hausdorff space. Let $x \in K^c$. For all $y \in K$, $\exists\, U_y, V_y$ open such that $U_y \cap V_y = \varnothing$,

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$x \in V_y$, $y \in U_y$. $\{U_y\}_{y\in K}$ is an open cover of K, hence there is a finite subcover $\{U_{y_i}\}_{i=1}^n$. Then take the intersection of the corresponding V_{y_i}:

$$\Big(\bigcap_{i=1}^n V_{y_i}\Big) \cap \Big(\bigcup_{i=1}^n U_{y_i}\Big) = \varnothing. \qquad \text{Then} \qquad \bigcup_{i=1}^n U_{y_i} \supseteq K \implies \Big(\bigcap_{i=1}^n V_{y_i}\Big) \cap K = \varnothing.$$

Every $x \in K^c$ has such an open neighbourhood $\bigcap_{i=1}^n V_{y_i}$ about x in K^c, hence K^c is a union of open sets $\implies$ K^c open $\implies$ K closed.



Tychonoff's theorem stands alone, and its proof doesn't justify our use of it throughout the course, hence the proof is left out (Tychonoff is $\iff$ the Axiom of Choice). □

Example 4.8

(a) $[0,1]$ and $2^{\mathbb{N}}$. They are tightly connected. On one hand, $2^{\mathbb{N}}$ embeds into $[0,1]$ with its image being the $\frac{1}{3}$-Cantor set:

$$2^{\mathbb{N}} \hookrightarrow [0,1], \qquad x \longmapsto \sum_{n\in\mathbb{N}} 2 \cdot x(n) \cdot 3^{-n-1},$$

i.e. the ternary representation of the Cantor set; i.e. at each interval $[a,b]$: 0 = recurse left, 2 = recurse right. (A different approach/formulation is provided in HW 1.) On the other

hand, $2^{\mathbb{N}}$ continuously surjects onto $[0,1]$ by

$$2^{\mathbb{N}} \twoheadrightarrow [0,1], \qquad x \longmapsto \sum_{n\in\mathbb{N}} x(n)\cdot 2^{-n-1}.$$

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The latter is not injective since in binary $0.0111\ldots = 0.100\ldots$. We may think of the Cantor space as the "most disconnected" (unraveled) version of $[0,1]$, as we distinguish in the Cantor space $0111\ldots$ from $1000\ldots$; and $2^{\mathbb{N}}$ is disconnected: taking $[0]_2$ and $[1]_2$ gives two disjoint open sets whose union is the entire $2^{\mathbb{N}}$.

(b) $\mathbb{T} := \mathbb{R}/\mathbb{Z} \cong S^1 \subseteq \mathbb{C}$, the torus* (see §4.4 on the torus and quotient topology at the end of this section).

(c) The Hilbert cube $[0,1]^{\mathbb{N}}$. □

4.2 Universal Compact Metrizable Spaces

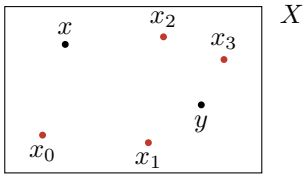
Theorem 4.9

Every separable metrizable topological space embeds into the Hilbert cube $[0,1]^{\mathbb{N}}$. In particular, Polish spaces are, up to homeomorphism, the G_δ subsets of $[0,1]^{\mathbb{N}}$; and compact Polish spaces (= compact metrizable) are, up to homeomorphism, the closed subsets of $[0,1]^{\mathbb{N}}$.

Proof. Let $D = \{x_n\}_n \subseteq X$ be a countable dense set. Let $d \leq 1$ be a compatible (truncated) metric on X (lecture 1, Proposition 2.2). Define $f : X \to [0,1]^{\mathbb{N}}$ by

$$x \longmapsto \{d(x,x_n)\}_{n\in\mathbb{N}},$$

where every point in X is encoded by its position relative to the "poles" $\{x_n\}_n$.



Injective because for $x \neq y \in X$, let $\varepsilon = d(x,y)$; then $B_{\varepsilon/2}(x)$ contains some $x_n \in D$ by density. Then $d(x,x_n) < \varepsilon/2$ but $d(y,x_n) \geq \varepsilon/2$. Continuous since for each n, $x \mapsto d(x,x_n)$

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is continuous. [For the continuity of f^{-1}, suppose in the image $d(y_k,x_n) \to d(y,x_n)$ for all $n \in \mathbb{N}$. Pick some x_n within $\varepsilon/2$ of y. Then eventually $d(y_k,x_n) < \varepsilon/2$, so $d(y_k,y) < \varepsilon$ by the triangle inequality.]

G_δ: Complete metrizability is a topological property (i.e. carried over by homeomorphisms)*. Indeed, if X admits a complete compatible metric d_X and f is a homeomorphism, then we can define the pullback metric on $f(X)$ via $d(f(x),f(y)) := d_X(x,y)$. Then this makes (X,d_X) isometric to $(f(X),d)$, and completeness passes through by isometry. Furthermore, d is compatible with the topology on $f(X)$, since the topology induced by d on $f(X)$ $\longleftrightarrow$ the topology induced by d_X on X (by definition of d) $\longleftrightarrow$ the topology on $f(X)$ (by the fact that f is a homeomorphism). Thus $(f(X),d)$ is a Polish

subspace of $[0,1]^{\mathbb{N}}$, since complete metrizability and second countability are both topological properties inherited from X. $f(X) \subseteq [0,1]^{\mathbb{N}}$ is Polish with the subspace topology $\implies$ $f(X)$ is G_δ (by the theorem characterizing Polish subspaces, Theorem 2.9).

Closedness for compact Polish spaces: Compactness is a topological property. Hence $f(X)$ is compact $\implies$ $f(X)$ is closed (and bounded) in a metric space. □

Remark 4.10 (∗)

Completeness itself is not a topological property, trivially: completeness requires a metric to describe Cauchyness, while a mere topological space may not even be metrizable. □

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It will be better for us to encode all Polish spaces as closed subsets of some Polish space Y, because the set $\mathcal{F}(Y) = \{\text{all closed subsets of } Y\}$ admits a Polish topology itself; thus we can think of $\mathcal{F}(Y)$ as the Polish space of all Polish spaces.

Theorem 4.11

Every Polish space is homeomorphic to a closed subset of $\mathbb{R}^{\mathbb{N}}$. Hence the set $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$ "=" {all Polish spaces}.

Proof. Given a Polish space X, we can identify X with a G_δ subset of $[0,1]^{\mathbb{N}}$ (Theorem 4.9). As before (in the proof of Theorem 2.9 that G_δ subsets are Polish), we embed X into $X \times \mathbb{R}^{\mathbb{N}} \subseteq \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}} \cong \mathbb{R}^{\mathbb{N}}$ (splitting the coordinates into odd ⊔ even gives this homeomorphism) as a closed subset: Let $X = \bigcap_{n\in\mathbb{N}} U_n$, fix a complete compatible metric $d \leq 1$ on $\mathbb{R}^{\mathbb{N}}$ (or on $[0,1]^{\mathbb{N}}$, since it is closed/compact in $\mathbb{R}^{\mathbb{N}}$) and define $f : X \to \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$ by

$$x \longmapsto \left(x,\ (d(x, U_n^c)^{-1})_{n\in\mathbb{N}}\right).$$

By the same argument as before, the second coordinate ensures that $f(X)$ is closed in $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$: convergence in $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$ means convergence in each component, hence $x_m \to x$ and $d(x_m, U_n^c)^{-1} \to r_n$. Hence $d(x_m, U_n^c)^{-1} \leq M_n$ for some $M_n > 0$ $\implies$ $d(x_m, U_n^c) \geq M_n^{-1} > 0$. Hence by continuity of the metric, $d(\,\cdot\,, U_n^c) : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}$ is continuous, and since $x_m \xrightarrow{d} x$, $d(x_m, U_n^c) \to d(x, U_n^c)$. Thus $d(x, U_n^c) \geq M_n^{-1} > 0$, and thus by uniqueness of limits, $r_n = \lim_{m\to\infty} d(x_m, U_n^c)^{-1} = d(x, U_n^c)^{-1} > 0$.

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Therefore $f(x_m) \to (x, (d(x, U_n^c)^{-1})_{n\in\mathbb{N}}) = f(x) \in f(X)$. And as before, f is continuous, as $x \mapsto x$ and $x \mapsto d(x, U_n^c)^{-1}$ are continuous, and convergence $x_m \to x$ in X $\implies$ $x_m \to x$ and $d(x_m, U_n^c)^{-1}$ converge $\implies$ $\{(x_m, (d(x_m, U_n^c)^{-1})_{n\in\mathbb{N}})\}_m$ converges in $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$ as $m \to \infty$, hence f is continuous. f^{-1} is continuous by definition of the product topology. □

Swapping the direction, we give a universal space with respect to surjections.

Theorem 4.12

Every nonempty compact metrizable space is a continuous image of the Cantor space $2^{\mathbb{N}}$.

Remark 4.13

If this were true with $[0,1]$ instead of $2^{\mathbb{N}}$, we'd say that every nonempty compact metrizable space is a curve. It is not, but remember that $2^{\mathbb{N}}$ is tightly related to $[0,1]$, being the "most disconnected" $[0,1]$, so with $2^{\mathbb{N}}$ we can consider every such space to be a "disconnected curve". □

Proof. Since every nonempty compact metrizable space is homeomorphic to a closed subset of $[0,1]^{\mathbb{N}}$, we first show this for $[0,1]^{\mathbb{N}}$.

But we also have homeomorphisms $2^{\mathbb{N}} \cong 2^{\mathbb{N}\times\mathbb{N}} \cong (2^{\mathbb{N}})^{\mathbb{N}}$:

$2^{\mathbb{N}} \cong 2^{\mathbb{N}\times\mathbb{N}}$: By the bijection $\mathbb{N}\times\mathbb{N} \xrightarrow{\sim} \mathbb{N}$ (the same map as in the cardinality argument): let f be the bijection induced by this map. Then X is open in

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$2^{\mathbb{N}\times\mathbb{N}} \iff$ finite membership condition (a finite subset of $\mathbb{N}\times\mathbb{N}$ specifying the conditioned indices) $\iff$ finite membership condition in $\mathbb{N} \iff f(X)$ is open in $2^{\mathbb{N}}$.

$2^{\mathbb{N}\times\mathbb{N}} \cong (2^{\mathbb{N}})^{\mathbb{N}}$: as both sets consist of two-dimensional arrays. For $f : \mathbb{N}\times\mathbb{N} \to 2$, we define $\tilde{f} : \mathbb{N} \to 2^{\mathbb{N}}$ via $i \mapsto f_i : \mathbb{N} \to 2$, $j \mapsto f(a_{ij})$:

$$\begin{array}{cccc} a_{11}, & a_{12}, & a_{13}, & \cdots \\ a_{21}, & a_{22}, & a_{23}, & \cdots \\ a_{31}, & a_{32}, & a_{33}, & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{array} \qquad \begin{array}{l} \text{viewed horizontally} = (2^{\mathbb{N}})^{\mathbb{N}}, \\ \text{together} = 2^{\mathbb{N}\times\mathbb{N}}. \end{array}$$

The map $\varphi : 2^{\mathbb{N}\times\mathbb{N}} \to (2^{\mathbb{N}})^{\mathbb{N}}$, $f \mapsto \tilde{f}$, is a homeomorphism: by topology (exercise), to show continuity it suffices to show continuity in each component:

$$\begin{aligned} \varphi \text{ continuous} &\iff f \mapsto \varphi(f)(n) \in 2^{\mathbb{N}} \text{ is continuous} \\ &\iff f \mapsto (\varphi(f)(n))(m) = f(n,m) \text{ is continuous} \end{aligned}$$

(applying the topology result again to this map into the product space $2^{\mathbb{N}}$; the previous one applies to $(2^{\mathbb{N}})^{\mathbb{N}}$). This map is continuous since it is the canonical projection $\pi_{(n,m)}$, which is continuous by definition of the product topology. φ is a continuous bijection, $2^{\mathbb{N}\times\mathbb{N}}$ is compact by Tychonoff, $(2^{\mathbb{N}})^{\mathbb{N}}$ is Hausdorff (metrizable, since it is a product of the Polish spaces $2^{\mathbb{N}}$); thus by part (e) of the theorem above, φ is a homeomorphism.

Let $\pi : 2^{\mathbb{N}} \to [0,1]$ be the continuous surjection in the last example (Example 4.8); then $\pi^{\mathbb{N}} : (2^{\mathbb{N}})^{\mathbb{N}} \to [0,1]^{\mathbb{N}}$ is defined by

$$(x_n)_n \longmapsto \big(\pi(x_n)\big)_n, \qquad x_n \in 2^{\mathbb{N}}.$$

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Hence $2^{\mathbb{N}} \xrightarrow{\cong} (2^{\mathbb{N}})^{\mathbb{N}} \to [0,1]^{\mathbb{N}}$ is the desired continuous surjection from $2^{\mathbb{N}}$ onto $[0,1]^{\mathbb{N}}$.

Let X be an arbitrary nonempty compact metrizable space. We identify X with a closed subset of $[0,1]^{\mathbb{N}}$ (Theorem 4.9). Recall the proposition that every nonempty closed subset C of $A^{\mathbb{N}}$ is a retract of $A^{\mathbb{N}}$ (Proposition 3.17), i.e. there is a continuous map from $A^{\mathbb{N}}$ onto C. Then $f^{-1}(X) \subseteq 2^{\mathbb{N}}$ is closed, hence a retract of $2^{\mathbb{N}}$ by a retraction map g. Hence $f \circ g : 2^{\mathbb{N}} \twoheadrightarrow f^{-1}(X) \twoheadrightarrow X$ is the continuous surjection we want. □

4.3 The Hyperspace of Compact Subsets

Points = compact subsets.

Definition 4.14

Let X be a topological space. Denote by $\mathcal{K}(X)$ the set of its compact subsets. We equip $\mathcal{K}(X)$ with the Vietoris topology, the one generated by the sets of the following two forms: for all $U \subseteq X$ open,

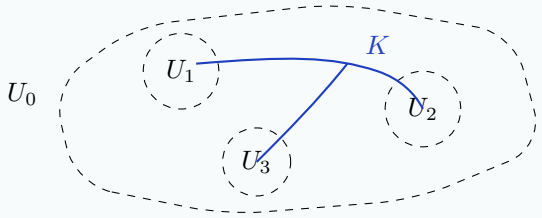
$$\left.\begin{aligned}(U)_{\mathcal{K}} &:= \{K \in \mathcal{K}(X) : K \subseteq U\},\\ [U]_{\mathcal{K}} &:= \{K \in \mathcal{K}(X) : K \cap U \neq \varnothing\}\end{aligned}\right\} \text{ subbasis.}$$ □

Fact 4.15

Because $(U_0)_{\mathcal{K}} \cap (U_1)_{\mathcal{K}} = (U_0 \cap U_1)_{\mathcal{K}}$, the following sets form a basis:

$$\langle U_0; U_1, U_2, \dots, U_n\rangle_{\mathcal{K}} := (U_0)_{\mathcal{K}} \cap [U_1]_{\mathcal{K}} \cap \cdots \cap [U_n]_{\mathcal{K}},$$

where $U_i \subseteq X$ is open for each i and $U_1, \dots, U_n \subseteq U_0$.



Indeed, $\langle U_0; U_i\rangle_{\mathcal{K}}$ is a finite intersection in the Vietoris topology;

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meanwhile, every finite intersection $\left(\bigcap_i [W_i]_{\mathcal{K}}\right) \cap \left(\bigcap_j (V_j)_{\mathcal{K}}\right)$ with $W_i, V_j \subseteq X$ open sets can be rewritten into that form: let $U_0 := \bigcap_j V_j$ and $U_i := W_i \cap U_0$. Then by definition of $(\cdot)_{\mathcal{K}}$, $[\cdot]_{\mathcal{K}}$,

$$\left(\bigcap_i [W_i]_{\mathcal{K}}\right) \cap \left(\bigcap_j (V_j)_{\mathcal{K}}\right) = \langle U_0; U_i\rangle_{\mathcal{K}}.$$

Thus every open set can be written as a union of these sets, hence they form a basis of the Vietoris topology. □

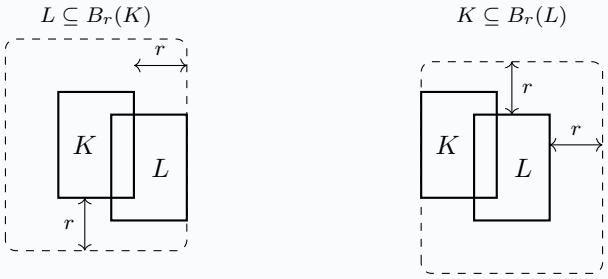
We can show that if X is metrizable, then so is $\mathcal{K}(X)$; in fact, for any compatible metric d on X:

Definition 4.16

Given a compatible metric $d \leq 1$ on X, we define a compatible metric d_H on $\mathcal{K}(X)$, called the Hausdorff distance, as follows: for $K, L \in \mathcal{K}(X)$,

$$d_H(K, L) := \begin{cases}\inf\{r \geq 0 : L \subseteq B_r(K) \text{ and } K \subseteq B_r(L)\}, & \text{if } K, L \neq \varnothing,\\ 1, & \text{otherwise,}\end{cases}$$

i.e. the smallest r such that L is contained in the puffed version of K and vice versa.



Remark.

$$d_H(K,L) = \begin{cases} \max\left\{ \max\limits_{x\in K} d(x,L),\ \max\limits_{y\in L} d(y,K) \right\}, & \text{if } K, L \neq \varnothing, \\ 1, & \text{otherwise.} \end{cases}$$

□

Proposition 4.17

d_H is compatible with the Vietoris topology.

Proof. Exercise (homework).

Hint: It suffices to show: ① the subbasic sets $(U)_{\mathcal{K}}$ and $[U]_{\mathcal{K}}$ of the Vietoris topology are open with respect to d_H, by showing

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that $\forall K \in (U)_{\mathcal{K}}$ (respectively $[U]_{\mathcal{K}}$), $\exists\, \varepsilon > 0$ such that $B_\varepsilon^{d_H}(K) \subseteq (U)_{\mathcal{K}}$ (respectively $[U]_{\mathcal{K}}$), using the double max statement. ② And show that the basic open d_H-balls are open in the Vietoris topology, by showing that $\forall K \in \mathcal{K}(X)$, $\varepsilon > 0$, there exists $\mathcal{C} \subseteq \mathcal{K}(X)$ Vietoris open such that

$$K \in \mathcal{C} \subseteq B_\varepsilon^{d_H}(K).$$

To prove this, observe that K can be covered by open balls of radius $\varepsilon/2$, $\{U_1, \dots, U_n\}$, $U_i = B_{\varepsilon/2}^{d_X}(x_i)$, $x_i \in K$. Then prove that

$$\mathcal{C} := \left\langle B_{\varepsilon/2}^{d_X}(K);\, U_1, \dots, U_n \right\rangle_{\mathcal{K}(X)}$$

satisfies the conditions above: $\left(B_{\varepsilon/2}^{d_X}(K)\right)_{\mathcal{K}}$ bounds $\max_{y\in L} d(y,K)$; $[U_i]_{\mathcal{K}}$ bounds $\max_{x\in K} d(x,L)$. □

Corollary 4.18

If X is metrizable, so is $\mathcal{K}(X)$ equipped with the Vietoris topology (with the Hausdorff metric).

Theorem 4.19

Let X be a topological space.

(a) If X has a metric d, then d_H metrizes $\mathcal{K}(X)$.
(b) If X is separable, then so is $\mathcal{K}(X)$.

(c) If X has a complete metric d, then $(\mathcal{K}(X), d_H)$ is complete.
(d) If X is Polish, then so is $\mathcal{K}(X)$.
(e) If X is compact metrizable, then so is $\mathcal{K}(X)$.

Proof. (a) Homework (outlined above, in the hint for Proposition 4.17).

(b) Let $D \subseteq X$ be a countable dense set. Then let $[D]^{<\infty} := \{S \subseteq D : S \text{ finite}\}$.

Claim. $[D]^{<\infty}$ is countable. One can show inductively that $[D]^n$ is countable, since every subset

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of size n can be viewed as an element of $[D]^{n-1}$ unioned with an element of D. Thus $[D]^{<\infty} = \bigsqcup_{n\in\mathbb{N}}[D]^n$ is a countable union of countable sets. Alternatively, enumerate $D = \{a_k\}_k$ and order the prime numbers $\{p_1 = 2, p_2 = 3, p_3 = 5, \dots\}$. For a finite subset $D' \subseteq D$, define the injection

$$\iota : [D]^{<\infty} \hookrightarrow \mathbb{N}, \qquad D' \longmapsto \prod_{a_i \in D'} p_i$$

(e.g. $\{a_1, a_2\} \mapsto 2 \cdot 3 = 6$). Thus $\aleph_0 \le |[D]^{<\infty}| \le \aleph_0 \implies |[D]^{<\infty}| = \aleph_0 = |\mathbb{N}|$.

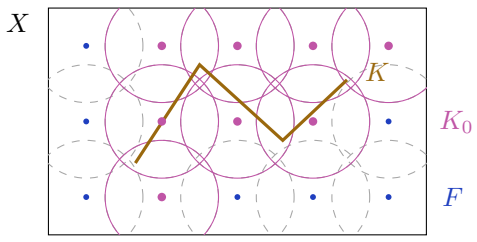
$[D]^{<\infty} \subseteq \mathcal{K}(X)$ since all finite sets are compact. $[D]^{<\infty}$ is dense: let $\langle U_0; U_1, \dots, U_n\rangle$ be a basic open set with $U_1, \dots, U_n \subseteq U_0$. Then since D is dense in a topological space, every nonempty open subset intersects D. Pick $d_i \in U_i \cap D$; then $\{d_1, \dots, d_n\} \in \langle U_0; U_1, \dots, U_n\rangle \cap [D]^{<\infty}$.

(c) We omit the proof. Challenge. See the online notes. Long.

(d) follows from (b), (c).

(e) Suppose X is compact metrizable; then every compatible metric is complete. Fix one metric d; then d_H is a complete metric on $\mathcal{K}(X)$. Recall that to show compactness, it remains to show total boundedness with respect to d_H (Theorem 4.2(c)): fix $\varepsilon > 0$ and a finite set $F \subseteq X$ that is ε-dense in X, i.e. $X = \bigcup_{x\in F} B_\varepsilon(x)$ (it exists since X is compact).

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Then $\mathcal{P}(F)$ is (2ε)-dense in $\mathcal{K}(X)$: for any $K \in \mathcal{K}(X)$, take

$$K_0 := \{x \in F : B_\varepsilon(x) \cap K \neq \varnothing\};$$

then $K \subseteq \bigcup_{x\in K_0} B^d_\varepsilon(x)$ and $K_0 \subseteq B^d_\varepsilon(K)$, thus $d_H(K, K_0) \le \varepsilon < 2\varepsilon$. Since it holds for all $\varepsilon > 0$, it follows that $\mathcal{K}(X)$ is totally bounded, hence compact. □

4.4 Torus and Quotient Topology

(∗) The following is excerpted from my MATH 570 notes, thus the explanations may be intended for something specific, yet it should briefly introduce the notion of a torus.

Definition 4.20

For a surjection $q : X \to Y$, we say $U \subseteq Y$ is open if $q^{-1}(U)$ is open in X. The topology of these open sets is called the quotient topology on Y. □

Define $x \sim y :\iff q(x) = q(y)$. On the other hand, if $\sim$ is a given equivalence relation, we define $q : X \to X/\sim$ via $q(x) = [x]$.

Lemma 4.21

A map out of a quotient $f : X/\sim \to Z$ is continuous $\iff$ $f \circ q$ is continuous.

Lemma 4.22

A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Proposition 4.23

Viewing $\mathbb{Z}^2$ as a normal subgroup of $\mathbb{R}^2$, the quotient $\mathbb{R}^2/\mathbb{Z}^2$ with the quotient topology is homeomorphic to $S^1 \times S^1$, where $S^1 = \{x + iy : x^2 + y^2 = 1\} \subseteq \mathbb{C}$.

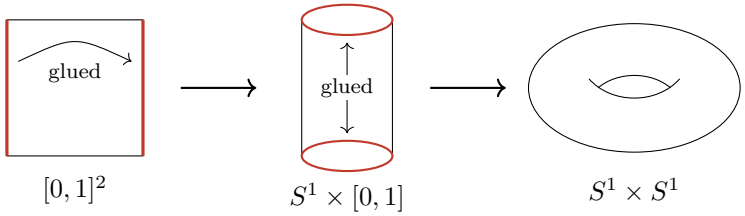
Proof. Define $F : \mathbb{R}^2 \to S^1 \times S^1$ via

$$(x, y) \longmapsto (e^{2\pi i x}, e^{2\pi i y});$$

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this map is surjective and continuous. By Lemma 4.21, the induced $f : \mathbb{R}^2/\sim \hookrightarrow\!\!\!\!\rightarrow S^1 \times S^1$ is continuous. $S^1 \times S^1$ is a metric space, hence Hausdorff; notice that the quotient map $q : \mathbb{R}^2 \twoheadrightarrow \mathbb{R}^2/\mathbb{Z}^2$ is continuous by definition of the quotient topology, hence $\mathbb{R}^2/\mathbb{Z}^2 = q([0,1]^2)$, i.e. a continuous image of a compact set, hence compact. Hence by Lemma 4.22, f is a homeomorphism. Lastly, notice that $\mathbb{R}^2/\sim = \mathbb{R}^2/\mathbb{Z}^2$ as sets, hence they are equipped with the same quotient topology. Thus $\mathbb{R}^2/\mathbb{Z}^2 \cong S^1 \times S^1$. □

Remark. One way to realize this:



Indeed, one can show that $[0,1]^2/\sim$ is homeomorphic to $\mathbb{R}^2/\mathbb{Z}^2$. Applying twice the pair of lemmas (Lemmas 4.21 and 4.22) to the two surjections above, we have

$$\mathbb{R}^2/\mathbb{Z}^2 \;\cong\; [0,1]^2/\sim \;\cong\; S^1 \times S^1. \qquad \square$$

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5 Perfect Polish Spaces

Recall. In a topological space X, a point $x \in X$ is called isolated if $\{x\}$ is open; otherwise, it is called an accumulation / limit / cluster point. A topological space X is called perfect if it has no isolated points.

Example 5.1

$\mathbb{Q}$, $\mathbb{R}$, $2^{\mathbb{N}}$, $\mathbb{R}^{\mathbb{N}}$, $[0,1]^{\mathbb{N}}$, $\mathbb{N}^{\mathbb{N}}$ are perfect. $\square$

Definition 5.2

A subset Y of a topological space X is called a perfect subset ($\neq$ perfect) if it is closed and perfect with the relative/subspace topology. $\square$

Example 5.3

(a) $\mathbb{Q}$ is a perfect space but not a perfect subset of $\mathbb{R}$.
(b) $X = (0,1) \cup [2,3]$. $Y = (0,1) \cup \{2\}$ is not a perfect subset of X, but $Z = (0,1)$ is. (Y is not perfect on its own.) $\square$

Theorem 5.4 (Cantor)

Every nonempty perfect Polish space contains a homeomorphic copy of $2^{\mathbb{N}}$.

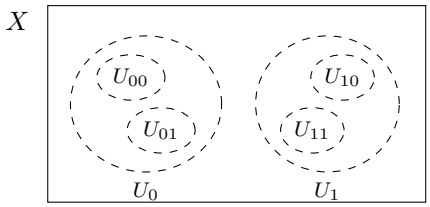
Proof. Let X be nonempty perfect. We build an embedding via a Cantor scheme (Definition 3.20). Namely, define $(U_s)_{s \in 2^{<\mathbb{N}}}$ satisfying, for all $s \in 2^{<\mathbb{N}}$:

(i) $U_s \neq \varnothing$ is open, and $U_{si} \cap U_{sj} = \varnothing$ for $i \neq j$;
(ii) $\operatorname{diam}(U_s) < \dfrac{1}{|s|+1}$;
(iii) $\overline{U_{si}} \subseteq U_s$ for all $i \in \{0,1\}$,

where the diameter is taken with respect to a complete compatible metric $d \leq 1$. Granted such a scheme, the associated (injective continuous, Proposition 3.19) function f has domain $D = 2^{\mathbb{N}}$,

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hence is an embedding of $2^{\mathbb{N}}$ into X (by Proposition 3.19 on Luzin schemes).



We build the scheme inductively: put $U_\varnothing := X$. Then, given nonempty open U_s, $s \in 2^{<\mathbb{N}}$, because there is no isolated point, there exist distinct $x, y \in U_s$. Then $\exists\, \varepsilon \in (0, \frac{1}{|s|+2})$ such that $B_\varepsilon(x) \cap B_\varepsilon(y) = \varnothing$, and $\overline{B_\varepsilon(x)}, \overline{B_\varepsilon(y)} \subseteq U_s$. Put $U_{s0} := B_\varepsilon(x)$ and $U_{s1} := B_\varepsilon(y)$. □

Theorem 5.5

All Polish spaces have the perfect set property (PSP), i.e. every Polish space X is either countable or contains a nonempty (closed) perfect subset, hence a homeomorphic copy of $2^{\mathbb{N}}$ by Theorem 5.4. Thus the continuum hypothesis holds on the collection of Polish spaces.

This theorem is a corollary of the following.

Theorem 5.6 (Cantor–Bendixson)

Every Polish space X is a disjoint union $X = P \sqcup U$ of a perfect subset (closed) $P \subseteq X$ and a countable open $U \subseteq X$. The P is unique, hence so is U, and P is called the perfect kernel of X.

Proof. There are two proofs of existence, one good, one fast.

Fast proof: "σ-ideal argument". Let $\mathcal{B}$ be a countable basis. Let $\mathcal{U} \subseteq \mathcal{B}$ be the collection of countable basic open sets. Then consider the countable open set $U := \bigcup \mathcal{U}$. Then $P := X \setminus U$ is closed. We claim $X = P \sqcup U$ works.

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Fix $x \in P$ and a basic open neighbourhood $V \in \mathcal{B}$ of x. If $V \cap P = \{x\}$, then V is countable because $V = (U \cap V) \cup \{x\}$. Hence $V \in \mathcal{U}$, so $V \subseteq \bigcup \mathcal{U} = U$, contradicting $x \in V$ and $x \in P = X \setminus U$. Thus x is not isolated in P with the relative topology. The good proof is given later as a discussion.

Uniqueness: Suppose $P_1 \sqcup U_1 = X = P_2 \sqcup U_2$, where the P_i are perfect and the U_i are countable open. Then $P_1 \cap U_2$ and $P_2 \cap U_1$ are G_δ subsets of X (since closed sets are G_δ, Lemma 2.8), hence Polish (Theorem 2.9); and perfect, since an isolated point relative to the open set U_j is isolated in the whole space P_i. But P_i is perfect, thus $P_i \cap U_j$ has no isolated points. Thus $P_i \cap U_j$ is a perfect countable Polish space. By the proven Cantor's theorem (Theorem 5.4), $P_i \cap U_j = \varnothing$ (otherwise it contains $2^{\mathbb{N}}$, contradicting countability). Hence $P_1 \subseteq P_2 \sqcup U_2 = X$ with $P_1 \cap U_2 = \varnothing$, and $P_2 \subseteq P_1 \sqcup U_1 = X$ with $P_2 \cap U_1 = \varnothing \implies P_1 \subseteq P_2$ and $P_1 \supseteq P_2 \implies P_1 = P_2$, thus $U_1 = U_2$. □

Proof of the full Cantor's theorem (PSP, Theorem 5.5). By Cantor–Bendixson (Theorem 5.6), $X = P \sqcup U$ with U countable open and P a perfect subset (closed). Thus P is Polish. If $P = \varnothing$, then $X = U$ is countable. If $P \neq \varnothing$, then P contains a homeomorphic copy of $2^{\mathbb{N}}$ by Theorem 5.4. □

5.1 Towards the Good Proof of Existence

Good in the sense that this idea nurtures a lot

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of research in descriptive set theory.

Definition 5.7

For a topological space X, call $X' := X \setminus \{\text{isolated points}\}$ the Cantor–Bendixson derivative of X. □

Remark. Note that X' is closed, since each isolated point is open. Furthermore, if X is second countable, then $X \setminus X' = \{\text{isolated points of } X\}$ is countable (each of the isolated points must form a basic open set).

Definition 5.8 (HW 1)

An ordinal is a transitive set $\boxed{\text{well-ordered}}$ by membership $\in$. A transitive set means that every element of the set is also a subset of it; the ordering is given by $\alpha < \beta :\iff \alpha \in \beta$. Every element of an ordinal is itself an ordinal, and each ordinal consists exactly of all the ordinals smaller than it. There are two kinds of ordinals: the successor of an ordinal α is

$$\alpha + 1 = \underbrace{\alpha}_{\text{as a set}} \cup \underbrace{\{\alpha\}}_{\text{as an element of } \alpha+1},$$

which is the smallest ordinal greater than α. A limit ordinal is a nonzero ordinal that is not a successor. An ordinal is countable if it is countable as a set. The first uncountable ordinal, denoted ω_1, is the set of all countable ordinals. If A is a set of ordinals, we define the supremum of A as the minimal element among all upper bounds of A, denoted $\sup A$.

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(Existence by well-ordering.) Equivalently, $\sup A = \bigcup_{\alpha \in A} \alpha$. □

Remark 5.9

1. First, a few ordinals include the ones we have been working with without explicitly calling them so:

   $$0 := \varnothing, \quad 1 = \{\varnothing\} = \{0\}, \quad 2 = \{0, 1\}, \quad \ldots, \quad n = \{0, \ldots, n-1\}.$$

   Indeed $0 < 1$, $0 < 2$, $1 < 2$, $\ldots$: ordinals extend the ordering on $\mathbb{N}$, well-ordering the class of ordinals.
2. Ordinals that are not successors, i.e. limit ordinals, are precisely the difference from the ordering on $\mathbb{N}$. For instance, let F be the collection of all finite ordinals; then the first infinite ordinal is

   $$\omega := \mathbb{N} = \sup F = \bigcup_{\varphi \in F} \varphi = \bigcup_{n=0}^{\infty} n;$$

it is a limit ordinal, as it cannot be a successor: if an ordinal α satisfies $\alpha+1=\alpha\cup\{\alpha\}=\omega$, then $\alpha=n$ for some $n\in\omega$. Then $n+2\in\omega$ but $n+1=\omega\in n+2$, i.e. $n+2<\omega$ while $\omega<n+2$. This is impossible since $<$ is a well-ordering.

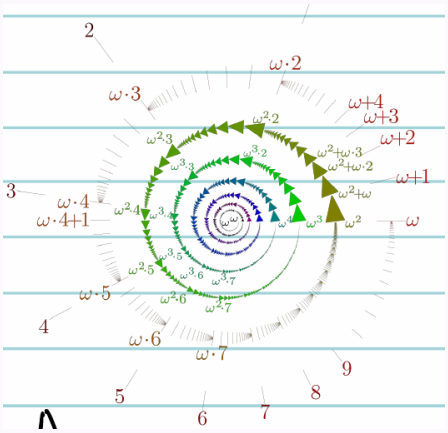
3. Limit ordinals are, equivalently, nonempty ordinals with no greatest element.
4. The first uncountable ordinal, denoted ω_1, is the supremum of all countable ordinals.
5. A sequence $(A_\alpha)_{\alpha<\gamma}$ is indexed by the elements of

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the well-ordered set γ.

6. Limit ordinals are the leaps between classes of successor ordinals:

$$0\to 1\to 2\to\cdots \dashrightarrow \omega\to\omega+1\to\omega+2\to\cdots$$
$$\dashrightarrow \omega\cdot 2=\omega+\omega\to\omega\cdot 2+1\to\omega\cdot 2+2\to\cdots$$
$$\dashrightarrow \omega\cdot 3\to\omega\cdot 3+1\to\cdots \dashrightarrow \cdots \dashrightarrow \underbrace{\omega^2}_{\sup_n(\omega\cdot n)}\to\cdots$$
$$\dashrightarrow \omega^3\to\cdots \dashrightarrow \underbrace{\omega^\omega}_{\sup_n(\omega^n)}\to\cdots$$



7. A sequence $(A_\alpha)_{\alpha<\gamma}$ is indexed by the elements of the well-ordered set γ.
8. Cardinality ($\aleph_0=|\mathbb{N}|$, $\aleph_1$, $2^{\aleph_0}=\mathfrak{c}$ the continuum) describes a different object than ordinality. Cardinality describes sizes; ordinality extends the well-ordering on $\mathbb{N}$.
9. Transfinite induction. Just like for $\mathbb{N}$, to show a statement is true for all $\gamma_0<\alpha<\gamma$, we prove ① the base case, ② the inductive step, for successor ordinals and limit ordinals *split into cases* (for $\mathbb{N}$, only the successor check is needed). □

Recall.

Proposition 5.10 (HW 1, Q1)

Let X be a second countable topological space. Then

(a) Every open set can be encoded as a subset of

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$\mathbb{N}$. Hence X has at most continuum-many open sets.

(b) Any strictly monotone sequence $(U_\alpha)_{\alpha<\gamma}$ of open sets in X has countable length, i.e. γ is countable. (Hint: note that the encoding in (a) is monotone, then produce an injection into $\mathbb{N}$.)

(c) Every monotone sequence $(U_\alpha)_{\alpha<\omega_1}$ of open subsets of X eventually stabilizes, i.e. $\exists \beta < \omega_1$ such that $\forall \beta \leq \alpha < \omega_1$, $U_\alpha = U_\beta$. (Use transfinite induction to build a contradiction towards (b).) Note that β is countable since ω_1 is the first uncountable ordinal.

(d) Taking complements, the same conclusion can be drawn for closed sets in X.

Definition 5.11

By transfinite induction, define

$$X^0 := X, \qquad X^{\alpha+1} := (X^\alpha)', \qquad X^\lambda := \bigcap_{\alpha<\lambda} X^\alpha \text{ if } \lambda \text{ is a limit ordinal.}$$

Thus $(X^\alpha)_{\alpha<\omega_1}$ is a decreasing sequence of closed sets. Since X is second countable (Polish), this sequence stabilizes at a countable $\gamma < \omega_1$ (Proposition 5.10). The least such ordinal γ is called the Cantor–Bendixson rank of X. We write $X^\infty := X^\gamma = X^\alpha$ for $\alpha \geq \gamma$. □

Theorem 5.12

Let X be Polish. For some countable ordinal $\alpha_0 < \omega_1$, we have $X^\alpha = X^{\alpha_0}$ for all $\alpha \geq \alpha_0$, and X^{α_0} is the perfect kernel of X.

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Proof. X^{α_0} cannot have isolated points: $X^{\alpha_0+1} = X^{\alpha_0}$. $X \setminus X^{\alpha_0}$ is countable open:

$$X \setminus X^{\alpha_0} = \bigcup_{\alpha<\alpha_0} (X^\alpha \setminus X^{\alpha+1}),$$

and each $X^\alpha \setminus X^{\alpha+1}$ is countable open. □

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6 Zero-Dimensional Spaces

Recall. A topological space X is disconnected if it admits a partition into two nonempty open sets, i.e. $\exists U_0, U_1 \subseteq X$ open such that $X = U_0 \sqcup U_1$ (hence U_0, U_1 are clopen). Equivalently, X is disconnected if it admits a nontrivial ($\neq \varnothing, X$) clopen set. Otherwise, call X connected. □

We briefly discuss the notion of connectedness, then give examples of it.

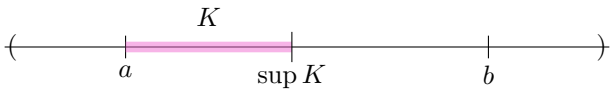
Definition 6.1 (Path-connectedness)

Let X be a topological space. A path from x to y in X is a continuous map $\gamma : [0,1] \to X$ with $\gamma(0) = x$, $\gamma(1) = y$. X is path-connected if $\forall x, y \in X$, $\exists$ a path γ from x to y. □

Theorem 6.2

Every interval of $\mathbb{R}$ is connected in the subspace topology. In particular, $\mathbb{R}$ and $[0,1]$ are connected. (We allow endpoints to be $\pm\infty$.)

Proof. Let I be an interval. Suppose $I = A \sqcup B$, nonempty and open.



Without loss of generality take $a < b$, $a \in A$, $b \in B$. Then $[a,b] \subseteq I$, since I is an interval. Let $K = A \cap [a,b]$. It is bounded above and contains a, thus $\sup K$ exists.

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In particular, $[a,b]$ is closed in $\mathbb{R}$, hence $\sup K \in [a,b]$. There is $(a_k)_{k\in\mathbb{N}} \subseteq K$ such that $a_k \to \sup K$ as $k \to \infty$. Thus $\sup K \in A$ (otherwise a_k eventually falls into B, which is impossible by disjointness). Denote $s = \sup K \in A$. Since A is open in I, by definition $\exists U \subseteq \mathbb{R}$ open such that $U \cap I = A$. Hence for $s \in A$, $\exists\, \delta > 0$ such that $B_\delta(s) \cap I \subseteq A$. Take $\varepsilon = \min\{\delta, b - s\}$. Then $(s, s+\varepsilon) \subseteq [s,b] \subseteq I$, and $(s,s+\varepsilon) \subseteq B_\delta(s) \implies (s,s+\varepsilon) \cap I = (s,s+\varepsilon) \subseteq A$. Then $s + \frac{\varepsilon}{2} \in [s,b] \subseteq [a,b]$, and $s + \frac{\varepsilon}{2} \in (s,s+\varepsilon) \subseteq A$ $\implies$ $s + \frac{\varepsilon}{2} \in K$. This is a contradiction since $s + \frac{\varepsilon}{2} > s = \sup K$. □

Proposition 6.3

If X is path-connected, then it is connected.

Proof. Suppose otherwise: $X = A \sqcup B$ with A, B nonempty open in X. Since X is path-connected, there is a continuous $\gamma : [0,1] \to X$ from a point of A to a point of B; thus

$$[0,1] = \gamma^{-1}(A \sqcup B) = \gamma^{-1}(A) \sqcup \gamma^{-1}(B)$$

is a disjoint union of two nonempty open sets. Yet $[0,1]$ is connected in the relative topology in $\mathbb{R}$ (Theorem 6.2). Contradiction. □

Theorem 6.4

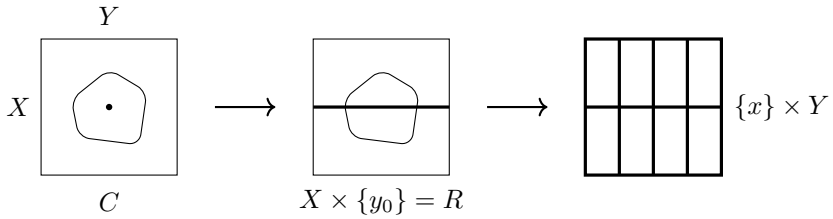
If X and Y are connected, then so is $X \times Y$. Thus every finite product of connected spaces is connected, by induction.

Proof. It suffices to show that a nonempty clopen set $C \subseteq X \times Y$ must be $X \times Y$. Take $(x_0, y_0) \in C$. $R = X \times \{y_0\}$ is a copy of X, i.e. homeomorphic

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to X by the canonical projection $\pi : (x, y_0) \mapsto x$. This is true since for an open set $U \subseteq X$, $U \times Y$ is open in $X \times Y$ (being a product of two open sets), and then $U \times \{y_0\} = (U \times Y) \cap (X \times \{y_0\})$ is open in $X \times \{y_0\}$ with respect to the relative topology. Thus $R = X \times \{y_0\}$ is connected (inherited from the connectedness of X). Then $C \cap R$ is clopen in R and contains (x_0, y_0), so $C \cap R = R$. Now take an

arbitrary $x \in X$; $(x, y_0) \in R \subseteq C$. Likewise, $\{x\} \times Y$ is connected, thus $(\{x\} \times Y) \cap C$ is clopen in $\{x\} \times Y$ and contains (x, y_0). Hence $(\{x\} \times Y) \cap C = \{x\} \times Y \implies \{x\} \times Y \subseteq C$ for all $x \in X \implies C = X \times Y$.



□

Theorem 6.5

An arbitrary product $\prod_i X_i$ of connected spaces is connected in the product topology.

Proof. If some X_i is empty, the product is empty: for a finite product this is obvious. For an arbitrary product, $\prod_i X_i = \{f : I \to \bigcup_i X_i \mid f(i) \in X_i\}$; if some $X_i = \varnothing$, then $\prod_i X_i = \varnothing$, since no function assigns i to "no element". Fix a point p of $X = \prod_i X_i$ (existence by the Axiom of Choice). Let D be the set of points of X that differ

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from p in only finitely many coordinates.

D is dense in X: Take a basic set U which restricts finitely many coordinates in $F \subseteq I$ to open subsets $U_i \subseteq X_i$, for these $i \in F$ (definition of the product topology). Take $u_i \in U_i$ for the $i \in F$. Define $x \in \prod_i X_i$ via

$$\begin{cases} x(i) = u_i, & \text{for } i \in F; \\ x(i) = p(i), & \text{for } i \notin F. \end{cases}$$

Then $x \in U \cap D \neq \varnothing$.

D is connected: Take a finite set F of indices, and let D_F be the set of all points agreeing with p on $i \notin F$. Then D_F is homeomorphic to $\prod_{i \in F} X_i$, which is a finite product of connected spaces, hence connected by Theorem 6.4. Moreover, $p \in D_F$ for all F, and

$$D = \bigcup_{\substack{F \subseteq I \\ \text{finite}}} D_F.$$

Let C be clopen in D. Replace C by $D \setminus C$ if $p \notin C$, so that $p \in C$. Then each $C \cap D_F$ is clopen in D_F with the relative topology, with $p \in C \cap D_F$. Each D_F is connected. Thus $C \cap D_F = D_F \implies D_F \subseteq C$. Hence $D_F \subseteq C$ for all finite $F \implies D = \bigcup D_F \subseteq C \implies D = C$.

X is connected: Connectedness extends from a dense subset: suppose $C \subseteq X$ is nonempty clopen. Then $C \cap D$ is

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clopen. Since D is dense, $C \cap D \neq \varnothing$. D is connected $\implies C \cap D = D \implies D \subseteq C$. C is closed and contains the dense set D, thus $\overline{D} = X \subseteq C$, since the closure is the smallest closed set containing the subset. Hence $X = C$, thus X is connected. □

Example 6.6

(a) $[0,1]^n$, $\mathbb{R}^n$, $\mathbb{C}^n$, $n \geq 1$, are connected. Note that $x, y \in \mathbb{K}^n$ can be connected by straight lines: $\gamma(t) = (1-t)x + ty$. Hence they are path-connected, hence connected. Alternatively, we showed that $\mathbb{R}$ is connected (Theorem 6.2), thus $\mathbb{R}^n$ is connected (Theorem 6.4), and $\mathbb{C} \cong \mathbb{R}^2$ and $\mathbb{C}^n \cong \mathbb{R}^{2n}$ are connected.

(b) More generally, every normed vector space $(V, \|\cdot\|)$ over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ is path-connected, hence connected by Proposition 6.3. Indeed, for $x, y \in V$ the straight-line path $\gamma(t) = (1-t)x+ty$, $t \in [0,1]$, is continuous (even Lipschitz), since

$$\|\gamma(t) - \gamma(s)\| = |t - s| \, \|y - x\|.$$

Nothing about dimension or separability is used, so this covers e.g. ℓ^p, $C([0,1])$ and ℓ^∞, and (a) is the special case $V = \mathbb{K}^n$. The same segment argument shows that every convex subset of V (e.g. a ball, or a linear subspace) is path-connected.

(c) $[0,1]^{\mathbb{N}}$, $\mathbb{R}^{\mathbb{N}}$, $\mathbb{C}^{\mathbb{N}}$ are connected. Convex interpolation $\gamma(t)$ still works fine, as we get continuity in each component, hence they are again path-connected. Alternatively, they are products of connected spaces (Theorem 6.5).

(d) A product of connected spaces is connected (proved: Theorems 6.4 and 6.5).

(e) $2^{\mathbb{N}}$, $\mathbb{N}^{\mathbb{N}}$, tree spaces ($A^{\mathbb{N}}$ with A equipped with the discrete topology) are disconnected. □

Definition 6.7

A topological space is 0-dimensional if it admits a basis of clopen sets. □

The 0-dimensional Polish spaces are "the most disconnected" spaces, to be formalized later.

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Remark 6.8

0-dimensional Hausdorff spaces are totally disconnected, i.e. the only connected subsets are singletons and $\varnothing$. The converse is not true, not even for Polish spaces. For example,

$$X := \{x \in \ell^1(\mathbb{N}) : x_n = 0 \text{ or } \tfrac{1}{n}\},$$

where $\ell^1(\mathbb{N}) := \{(x_k)_{k\in\mathbb{N}} : \sum_{k\in\mathbb{N}} |x_k| < \infty\}$ is complete with respect to the ℓ^1 norm $\|(x_k)_k\|_1 := \sum_{k\in\mathbb{N}} |x_k|$. X is closed in $\ell^1(\mathbb{N})$: if $(x^{(n)})_n$ is a Cauchy sequence in X, then $x^{(n)} \to x$ in $\ell^1(\mathbb{N})$. But Cauchy $\implies$ Cauchy in each component sequence $\implies (x_k^{(n)})_n \to x_k$ as $n \to \infty$, for every component $k \in \mathbb{N}$. But $x_k^{(n)} = 0$ or $\frac{1}{k}$, hence $x_k^{(n)} \to x_k$ implies the component stabilizes to 0 or $\frac{1}{k}$. Thus $x \in X$, hence X is closed.

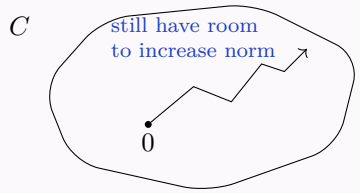
X is closed in a Polish space, hence X is Polish. It is obviously totally disconnected: take $Y \subseteq X$ with $|Y| \geq 2$. Pick $x \neq y \in Y$. Then x, y differ at some $n \in \mathbb{N}$. Define $U_0 := \{x' \in X : x'_n = x_n\}$,

$U_1 := \{x' \in X : x'_n = y_n\}$. They are easily clopen in X.

But X is not 0-dimensional: consider the unit ball at 0. There is no clopen set C such that $0 \in C \subseteq B_1(0)$: we claim elements in C have arbitrarily large norm. Suppose otherwise, that C is bounded by M; we work towards a contradiction. Imagine we walk starting from 0, $n = 1 \to 2 \to \cdots$. Switch coordinate n on $(0 \mapsto \frac{1}{n})$ if the point after switching is still in C (we

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increase the norm greedily at every step). Then we have a sequence $(x^{(n)})_n \subseteq C$.



We have skipped infinitely many coordinates, otherwise eventually the sum $\to \infty$, since it is the harmonic series minus finitely many terms. Moreover, $(x^{(n)})_n$ converges in ℓ^1 to some $x \in \ell^1(\mathbb{N})$, since $(x^{(n)})_n$ is Cauchy. Then $x \in C$, since C is closed.

Since C is open, $\exists \varepsilon > 0$ such that $B_\varepsilon(x) \subseteq C$. Pick $N \in \mathbb{N}$ such that $\sum_{i \geq N} x_i < \frac{\varepsilon}{2}$ (existence by absolute convergence). Pick $k > \max\{N, 2/\varepsilon\}$ that is *skipped* during the walk. Consider the element we would get if we took k: $y \in \ell^1(\mathbb{N})$ such that

$$y_n = \begin{cases} x_n, & \text{if } n < k, \\ \frac{1}{k}, & \text{if } n = k, \\ 0, & \text{if } n > k, \end{cases} \qquad \begin{bmatrix} x = x_1, \ldots, x_{k-1}, 0, x_{k+1}, \ldots \\ y = x_1, \ldots, x_{k-1}, \frac{1}{k}, 0, 0, \ldots \end{bmatrix}$$

Then

$$d_1(x, y) = \frac{1}{k} + \sum_{i \geq k} x_i < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $y \in B_\varepsilon(x) \subseteq C$. But this implies we should have switched coordinate k on, instead of skipping it. Contradiction. Thus C is unbounded, so $C \not\subseteq B_1(0)$, a contradiction. Such clopen C do not exist. Hence X is not 0-dimensional. □

Lemma 6.9

If X is a second countable space, then every basis $\mathcal{B}$ of X admits a countable $\mathcal{B}_0 \subseteq \mathcal{B}$ which is still a basis.

Proof. Recall from HW 1 that second countable spaces are Lindelöf, i.e. every open cover admits a countable subcover (this uses

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the axiom of (countable) choice). Fix a countable basis $\mathcal{U}$ of X, and for each $U \in \mathcal{U}$ observe that $\mathcal{B}_U := \{B \in \mathcal{B} : B \subseteq U\}$ is a cover of U, with $U = \bigcup \mathcal{B}_U$. Since U is second countable (inherited from its ambient space X), there is a countable cover $\mathcal{B}'_U \subseteq \mathcal{B}_U$ of U. Then $\mathcal{B}_0 := \bigcup_{U \in \mathcal{U}} \mathcal{B}'_U$ is a countable basis of X. □

Corollary 6.10

Zero-dimensional second countable spaces admit a countable basis of clopen sets.

6.1 Topological Characterization of $2^{\mathbb{N}}$ and $\mathbb{N}^{\mathbb{N}}$

Theorem 6.11 (Brouwer)

The Cantor space $2^{\mathbb{N}}$ is the unique (up to homeomorphism) nonempty zero-dimensional perfect compact metrizable ($\iff$ compact Polish) space.

Proof. Let X be such a space; we build a homeomorphism $2^{\mathbb{N}} \xrightarrow{\sim} X$. To do this we build a Cantor scheme (Definition 3.20) $(C_s)_{s\in 2^{<\mathbb{N}}}$ on X of vanishing diameter with respect to a fixed complete compatible metric d on X such that $\forall s \in 2^{<\mathbb{N}}$:

(i) $C_s \neq \varnothing$;
(ii) $C_\varnothing = X$ and $C_s = C_{s0} \sqcup C_{s1}$;
(iii) C_s is clopen.

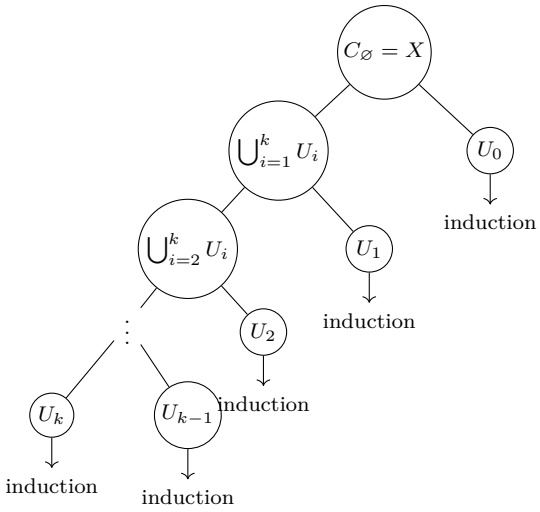
Given such a scheme, the associated function $f : D \to X$, $D \subseteq 2^{\mathbb{N}}$, would have domain D closed by (iii) and Proposition 3.19(d), and $D = 2^{\mathbb{N}}$ by (i) with the vanishing diameter. Thus f is

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a continuous injection from the compact $2^{\mathbb{N}}$ into the Hausdorff X, hence an embedding (Theorem 4.7(e)). Lastly, f is surjective by (ii), hence $2^{\mathbb{N}} \xrightarrow{\sim} X$.

Take $C_\varnothing := X$. By 0-dimensionality, there is a cover of X by clopen sets of small diameter $\leq \frac{1}{2}$, and by compactness there is a finite such cover $U_0, \dots, U_n$. Disjointify these, replacing each U_i by $U_i \setminus \bigcup_{j<i} U_j$; we get a finite partition $X = \bigsqcup_{i\leq n} U_i$ into clopen sets of diameter $\leq \frac{1}{2}$. We make sure that each $U_i \neq \varnothing$ by throwing away the empty sets. And we make sure $n \geq 1$: if U_0 is the only remaining one, then $U_0 = X$ and $|U_0| \geq 2$ since X is perfect. Take $x \neq y \in U_0$. By Hausdorffness ($\Leftarrow$ metrizable), $\exists\, V_1, V_2 \subseteq X$ open such that $V_1 \cap V_2 = \varnothing$, $x \in V_1$, $y \in V_2$. By 0-dimensionality, $\exists\, V$ clopen such that $x \in V \subseteq V_1$. Redefine $U_0 := V$, $U_1 := X \setminus V$, which are both clopen.

Call the obtained collection $U_0, \dots, U_k$. Define $C_{0^j 1} := U_j$ and $C_{0^k} := U_k$, as shown:



Then do the same now with the C_s leaves, with a shrinking diameter, and proceed by induction, noticing that each C_s is a nonempty 0-dimensional perfect compact metrizable space (in particular, perfectness (respectively compactness) is witnessed by open (respectively closed) subsets, hence

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C_s is perfect and compact since it is clopen in X). □

Theorem 6.12 (Alexandrov–Urysohn)

The Baire space $\mathbb{N}^{\mathbb{N}}$ is the unique (up to homeomorphism) nonempty 0-dimensional Polish space in which every compact set has empty interior (i.e. is closed nowhere dense).

Proof. HW 2. Use a Luzin scheme instead of a Cantor scheme and reproduce the previous proof (that of Theorem 6.11). Use the noncompactness condition to show that such a partition into countably many clopen sets exists. In particular, assuming A_s is defined, cover it by countably many clopen sets with diameter $< \frac{1}{n}$, then make the diameter small enough so that any such cover is necessarily infinite. (Hint: compact $\iff$ complete and totally bounded. Notice that every clopen set is not compact, thus not totally bounded. Then $\exists\, \delta_0 > 0$ for which there are no finite δ_0-covers.) □

Corollary 6.13

$\mathbb{R} \setminus \mathbb{Q} \cong \mathbb{N}^{\mathbb{N}}$.

Proof. Note that compact subsets of $\mathbb{R} \setminus \mathbb{Q}$ are closed nowhere dense, and $\mathbb{R} \setminus \mathbb{Q}$ is Polish since G_δ (Theorem 2.9), 0-dimensional, and nonempty. On the other hand, there is a nice explicit construction of a Luzin scheme on $\mathbb{R} \setminus \mathbb{Q}$ outlined in HW 1, and another continued-fraction proof of this homeomorphism, which I do not put here at all. □

6.2 Parameterization of Every Polish Space with $\mathbb{N}^{\mathbb{N}}$

Theorem 6.14

Every nonempty Polish space X is a continuous image of $\mathbb{N}^{\mathbb{N}}$. In fact, there is a closed $C \subseteq \mathbb{N}^{\mathbb{N}}$ and a

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continuous bijection $f : C \to X$ (not necessarily a homeomorphism).

Proof. The second part implies the first by taking $f \circ g$, where $g : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow C$ is a retraction (it exists since C is closed, Proposition 3.17). We build such an f as the associated map of a Luzin scheme (Definition 3.18, Proposition 3.19). We will define a scheme $(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ on X of vanishing diameter with respect to a compatible complete metric on X such that $\forall s \in \mathbb{N}^{<\mathbb{N}}$:

(i) $F_\varnothing = X$, $F_s = \bigsqcup_{i\in\mathbb{N}} F_{si}$;
(ii) $\overline{F_{si}} \subseteq F_s$ for all $i \in \mathbb{N}$.

Given such a scheme, the associated injective map $f : D \to X$ has closed domain by (ii) and is surjective by (i), hence is a continuous bijection from a closed subset of $\mathbb{N}^{\mathbb{N}}$ to X.

To build such a scheme by induction, we enforce one more condition:

(iii) F_s is F_σ.

Define $F_\varnothing := X$, and suppose F_s is defined and F_σ. Thus $F_s = \bigcup_{n\in\mathbb{N}} C_n$ with C_n closed. Fix $n \in \mathbb{N}$, cover C_n by countably many open sets of diameter $\leq \frac{1}{|s|+2}$ (doable since X is Polish), so $C_n = \bigcup_m (C_n \cap U_{nm})$, where each $U_{nm} \cap C_n$ is F_σ since it is (open $\cap$ closed) $= (F_\sigma \cap$ closed). Define $F_{nm} := C_n \cap U_{nm}$. Thus we get a cover of F_s as follows: $F_s = \bigcup_{n,m\in\mathbb{N}} F_{nm}$. Disjointifying them gives $F_s = \bigsqcup_{n,m} F'_{nm}$ with $F'_{nm} \subseteq F_{nm}$ still F_σ and of small diameter. But now

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$$\overline{F'_{nm}} \subseteq \overline{C_n \cap U_{nm}} \subseteq \overline{C_n} \subseteq C_n \subseteq F_s,$$

hence setting $(F_{si})_i = (F'_{nm})_{(n,m)\in\mathbb{N}\times\mathbb{N}}$ (enumerating the F'_{nm} as the children of F_s) finishes the proof. □

6.3 Closed Subsets of the Baire Space $\mathbb{N}^{\mathbb{N}}$

Theorem 6.15

Every 0-dimensional separable metrizable space X embeds into $\mathbb{N}^{\mathbb{N}}$ (hence into $2^{\mathbb{N}}$ by HW 1, Q5(b)). Every zero-dimensional Polish space is homeomorphic to a closed subset of $\mathbb{N}^{\mathbb{N}}$ (hence to a G_δ subset of $2^{\mathbb{N}}$, by Example 2.12).

Proof. For both, we construct a Luzin scheme $(C_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ on X with vanishing diameter such that $\forall s \in \mathbb{N}^{<\mathbb{N}}$,

(i) $C_\varnothing = X$, $C_s = \bigsqcup_{i\in\mathbb{N}} C_{si}$;
(ii) C_s is clopen (could be $\varnothing$).

Granted such a scheme, we immediately obtain the associated map $D \to X$ as an embedding (Proposition 3.19(c)). If additionally X is Polish, then X is complete, and $\overline{C_{si}} = C_{si} \subseteq C_s$ for all s, i boosts the domain D into a closed subset of $\mathbb{N}^\mathbb{N}$ (Proposition 3.19(d)).

To build such a scheme, define $C_\varnothing := X$. Given C_s defined, there exist clopen basic sets $\{K_n\}_{n\in\mathbb{N}}$ such that $C_s = \bigcup_n K_n$ and $\operatorname{diam}(K_n) \leq \frac{1}{|s|+2}$, which is possible by 0-dimensionality of X (Corollary 6.10). Disjointify them; the result is still a collection of clopen sets of diameter $\leq \frac{1}{|s|+2}$. Enumerate this disjoint collection as $(C_{si})_{i\in\mathbb{N}}$, the children of C_s. The scheme follows. □

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7 Locally Compact Hausdorff Spaces

This is not covered in lecture, but I am putting it here for the ones that are not totally familiar with general topology, as we will use it in the next section (§8).

Recall. Let X be a topological space and $x \in X$. Then $N \ni x$ is called a neighbourhood of x if $\exists U \subseteq X$ open such that $x \in U \subseteq N$; equivalently, if $x \in \operatorname{Int}(N)$.

Definition 7.1

A topological space X is locally compact if every point has a compact neighbourhood, i.e. $K \ni x$ such that $\exists U$ open with $x \in U \subseteq K$, and K compact in X. X is called locally compact Hausdorff if it is locally compact and Hausdorff (abbreviated LCH). A set $C \subseteq X$ is called precompact if its closure $\overline{C}$ is compact in X. □

Example 7.2

(i) $\mathbb{R}^n$, $\mathbb{C}^n$, finite-dimensional normed vector spaces over a complete field $\mathbb{K}$ are LCH by the Heine–Borel theorem for normed vector spaces (cf. Theorem 4.2).
(ii) Compact Hausdorff spaces are LCH, e.g. $2^\mathbb{N}$, $[0,1]^\mathbb{N}$.
(iii) Topological spaces equipped with the discrete topology are LCH.
(iv) $\mathbb{Q}$, $\mathbb{N}^\mathbb{N}$, $\mathbb{R}\setminus\mathbb{Q}$, $\mathbb{R}^\mathbb{N}$, infinite-dimensional normed vector spaces are non-LCH, since compact sets in these spaces have empty interior. In particular:
$\mathbb{N}^\mathbb{N}$: Take $C \subseteq \mathbb{N}^\mathbb{N}$ compact. Then C is closed ($\iff C = [T_C]$, from the section on trees, Proposition 3.7). By HW 1, Q6 (Proposition 3.11), $[T_C]$ is compact $\iff T_C$ is finitely branching. Thus it cannot

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contain any basic cylinder set $[s]_\mathbb{N}$, since every basic cylinder set branches infinitely.
$\mathbb{Q}$, $\mathbb{R}\setminus\mathbb{Q}$: Take a compact set $K \subseteq \mathbb{Q}$. Note that compactness of K depends only on K and the topology on K, not on the ambient space (as long as the topology is consistent with the relative topology). Thus $K \subseteq \mathbb{Q}$ is compact $\iff K$ is compact in its subspace topology and in $\mathbb{R}$. If K has nonempty interior, then $\exists (a,b) \cap \mathbb{Q} \subseteq K$. Since $\mathbb{Q}$ is dense in

$\mathbb{R}$, the closure $\overline{(a,b)\cap\mathbb{Q}} = [a,b]$. Since K is closed, $[a,b] \subseteq K$. Yet $K \subseteq \mathbb{Q}$. Contradiction. $\mathbb{R}\setminus\mathbb{Q}$ follows likewise (or alternatively, recall $\mathbb{R}\setminus\mathbb{Q} \cong \mathbb{N}^{\mathbb{N}}$ by HW 1).
$\mathbb{R}^{\mathbb{N}}$: Take $K \subseteq \mathbb{R}^{\mathbb{N}}$ compact. Let π_n be the canonical continuous (by definition of the product topology) projections. Then $\pi_n(K)$ is the continuous image of a compact set, hence compact in $\mathbb{R}$. Yet compactness $\implies$ bounded in $\mathbb{R}$, thus $K \subseteq \prod_{n\in\mathbb{N}}[-M_n, M_n]$ with $M_n \geq 0$ for each $n \in \mathbb{N}$. Then K cannot contain any basic open set, since every basic open set is free ($= \mathbb{R}$) in infinitely many coordinates.
Infinite-dimensional normed vector spaces: Recall that in MATH 255 we saw that in infinite-dimensional normed vector spaces, Heine–Borel fails. In particular, $\overline{B_1(0)}$ is not compact despite being closed and bounded. Take K compact, and suppose $\mathrm{Int}(K) \neq \varnothing$. Then $\exists$

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$B_\varepsilon(x) \subseteq K \implies B_{\varepsilon/2}[x] \subseteq K$ and is closed, hence compact (by sequential compactness). Rescaling $\varepsilon/2 \mapsto 2$ and shifting the space $y \mapsto y - x$ are homeomorphisms. Hence $B_2[0]$ is compact. Yet $B_1(0) \subseteq B_2[0]$ and $B_2[0]$ is closed $\implies \overline{B_1(0)} \subseteq B_2[0]$ is a closed subset of a compact set, hence compact. Contradiction. □

Observation. All spaces above except $\mathbb{Q}$ are Polish. Conversely, $[0,1]^{\mathbb{R}}$ is compact Hausdorff by Tychonoff's theorem (Theorem 4.7(g)) but not metrizable. Non-metrizability is unnecessary for the course, but to prove it: recall that a neighbourhood basis of $x \in X$ is a collection of open neighbourhoods $(B_i)_{i\in I}$ of x such that $\forall U \ni x$ open, $\exists\, B_i$ such that $x \in B_i \subseteq U$. Unlike a basis of a topological space X, where every open set is a union of the basic open sets, an open neighbourhood of x need not be a union of the neighbourhood-basic sets. A space is called first countable if every point $x \in X$ has a countable neighbourhood basis. All metric spaces are first countable. We show that $[0,1]^{\mathbb{R}}$ is not first countable, hence non-metrizable.

Suppose a point $p \in [0,1]^{\mathbb{R}}$ has a countable neighbourhood basis $(N_m)_{m\in\mathbb{N}}$ at p. The trick is to pass the neighbourhood basis to the basic open sets in $[0,1]^{\mathbb{R}}$. For every open N_m, $\exists$ a basic open set B_m in $[0,1]^{\mathbb{R}}$ such that $p \in B_m \subseteq N_m$. Then $(B_m)_{m\in\mathbb{N}}$ is also a neighbourhood basis at p. Every

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basic open set in $[0,1]^{\mathbb{R}}$ constrains only finitely many indices, say $F_m \subseteq \mathbb{R}$. Then $F = \bigcup_{m\in\mathbb{N}} F_m$ is countable, hence some $s \in \mathbb{R}$ is not constrained. Define

$$W := \{x \in [0,1]^{\mathbb{R}} : |x(s) - p(s)| < \tfrac{1}{2}\}.$$

W is an open (since it only constrains coordinate s) neighbourhood of p. Hence $\exists\, k \in \mathbb{N}$ such that $p \in B_k \subseteq W$. Set x such that

$$x(r) = \begin{cases} p(r), & r \neq s, \\ p(s) + \frac{1}{2}, & r = s \text{ and } p(s) \leq \frac{1}{2}, \\ p(s) - \frac{1}{2}, & r = s \text{ and } p(s) > \frac{1}{2}. \end{cases}$$

Then $x \notin W$, yet $x \in B_k$ since none of the B_m's constrains coordinate s, contradicting $B_k \subseteq W$.

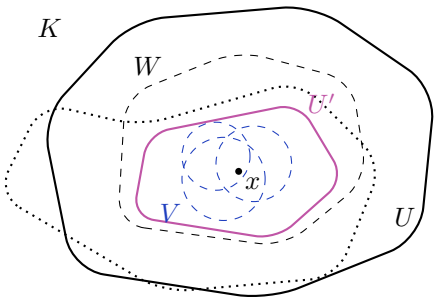
All that is to say, LCH does not imply metrizability, and vice versa. □

Notation. For a topological space X and $V, U \subseteq X$, we write $V \subseteq^* U :\iff \overline{V} \subseteq U$.

Lemma 7.3 (Shrinking lemma)

Let X be LCH, $U \subseteq X$ open, $x \in U$. Then $\exists V \ni x$ open with $\overline{V}$ compact and $\overline{V} \subseteq U$, i.e. $V \subseteq^* U$ and V precompact.

Proof. By local compactness, pick $K \ni x$ compact and $x \in W \subseteq K$ open. Then $U' := U \cap W \subseteq K$ is open, thus $C := K \setminus U'$ is closed in a compact set, hence compact. For each $c \in C$, separate it from x: pick $A_c \ni x$, $B_c \ni c$ open and disjoint (by Hausdorffness and the axiom of choice). $(B_c)_{c \in C}$ covers C, hence there is a finite subcover



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$\{B_{c_1}, \dots, B_{c_k}\}$ covering C. Let $A := \bigcap_{i=1}^k A_{c_i}$ and $B := \bigcup_{i=1}^k B_{c_i}$ (if $C = \varnothing$, pick $A = X$, $B = \varnothing$). Thus A is open, $A \cap B = \varnothing$, and $C \subseteq B$; consequently $A \cap C = \varnothing$.

Define $V := A \cap U'$, open. Then $\overline{V}$ is closed $\implies \overline{V} \subseteq K$, since K must contain all limit points of V. Then $\overline{V}$ is closed in the compact set K, hence compact. And since B is open, B^c is closed, thus $V \subseteq A \subseteq B^c$ implies $\overline{V} \subseteq B^c$ by closedness. Furthermore,

$$\overline{V} \subseteq B^c \cap K \subseteq C^c \cap K = (K \setminus U')^c \cap K = (K^c \cup U') \cap K = U' \cap K = U' \subseteq U. \qquad \square$$

Remark 7.4

In Hausdorff topological spaces, there are equivalent definitions of local compactness:

(1) Every point has a precompact neighbourhood.
(2) Every point has arbitrarily small compact neighbourhoods (small in the sense of inclusion). $\square$

7.1 LCH Spaces as Open Pieces of Compact Spaces

Definition 7.5

Let X be a topological space, $X^* := X \cup \{\infty\}$. Declare the open sets in X and the sets $\{\infty\} \cup (X \setminus K)$ with $K \subseteq X$ compact to be open in X^*. We call this X^* with this topology the one-point compactification of X. $\square$

Theorem 7.6

If X is LCH, then X^* is compact Hausdorff and X is an open subspace of X^*.

Proof (sketch). By Hausdorffness, compact subsets of X are closed in X, and finite unions and closed subsets of compact

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sets are compact. Hence the declared sets in Definition 7.5 give a topology.

Compact: Take any open cover of X^*. Then some open set contains ∞, hence is of the form $\{\infty\}\cup(X\setminus K)$. Then the remaining sets cover K. But K is compact, hence it must be covered by finitely many of them. Thus X^* is compact.

Hausdorff: Points in X are separated in X. To separate $x \in X$ from ∞, take $V \ni x$ with $\overline{V}$ compact in X (existence by local compactness). Then V and $\{\infty\} \cup (X \setminus \overline{V})$ separate x from ∞. □

Corollary 7.7

A topological space X is LCH $\iff$ it is homeomorphic to an open subspace of a compact Hausdorff space.

Proof. ($\Rightarrow$) Theorem 7.6. ($\Leftarrow$) Compact Hausdorff spaces are LCH. Open subsets of LCH spaces are LCH. In particular, local compactness is verbatim the shrinking lemma (Lemma 7.3). □

7.2 LCH and Polish Spaces

Proposition 7.8

For a topological space X, the following are equivalent:

(1) X is locally compact and Polish.
(2) X is LCH and second countable.

Such an X is σ-compact.

Proof. (1) $\Rightarrow$ (2) is clear from the definitions.

(2) $\Rightarrow$ (1): Fix a countable basis. Then its precompact members still form a basis: given $x \in U$ open,

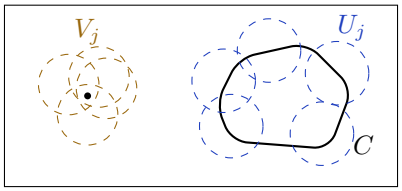
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there is a precompact open $V \ni x$ with $V \subseteq^* U$ by the shrinking lemma (Lemma 7.3). Then any basic $B \subseteq V$ with $x \in B$ is precompact, since $\overline{B}$ is a closed subset of the compact set $\overline{V}$. Enumerate them as $B_1, B_2, \ldots$. Define $K_n := \bigcup_{i=1}^{n} \overline{B_i}$.

Note that X is σ-compact, since $X = \bigcup_{n=1}^{\infty} \overline{B_n}$.

Polish: The sets B_n and $\{\infty\} \cup (X \setminus K_n)$ form a countable basis of X^*: take an open $\{\infty\} \cup (X \setminus K)$ in X^*. Then K compact in $X \implies K$ is covered by finitely many of the B_n's, thus $K \subseteq K_n$ for some n, hence $\{\infty\} \cup (X \setminus K_n) \subseteq \{\infty\} \cup (X \setminus K)$.

By Theorem 7.6, X^* is compact Hausdorff. By the same trick as in the proof of "closed subsets of compact spaces are compact" (Theorem 4.7) or the shrinking lemma (Lemma 7.3), one can prove that compact Hausdorff spaces are regular, i.e. a point and a disjoint closed set are separated by disjoint open sets. Thus by the Urysohn metrization theorem, X^* second countable regular Hausdorff $\implies$ metrizable. Thus X^* is compact metrizable $\iff$ compact Polish (Corollary 4.3). X is open in X^* by definition of X^*, hence X is Polish.



□

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8 Baire Category

(Despite the name, the theory of Baire category has nothing to do with the subject of category theory in algebra.)

8.1 Nowhere Dense Sets

Definition 8.1

In a topological space X, a set D is dense if it meets every nonempty open set, i.e. $\forall U \subseteq X$ nonempty open, $D \cap U \neq \varnothing$. We say a set $A \subseteq X$ is somewhere dense if $A \cap U$ is dense in U for some nonempty open set $U \subseteq X$. A is nowhere dense if it is not somewhere dense (abbreviated n.d.). □

Proposition 8.2

For a set A in a topological space, the following are equivalent:

(1) A is nowhere dense.
(2) In every nonempty open set U, there is a further nonempty open $V \subseteq U$ missed by A, i.e. $A \cap V = \varnothing$.
(3) $\overline{A}$ has empty interior.

Proof. (1) $\iff$ (2) is the negation of the definition of somewhere-density.

(2) $\Rightarrow$ (3): Suppose otherwise, $\operatorname{Int}(\overline{A}) \neq \varnothing$. Then $\exists\, x \in U \subseteq \overline{A}$ such that U is open. Claim: $A \cap U$ is dense in U, hence contradicting A being nowhere dense. Pick $\varnothing \neq V \subseteq U$ open. Then $V \subseteq U \subseteq \overline{A}$ $\implies$ $\exists\, y \in V \subseteq \overline{A}$ such that $y \in A \cap V \neq \varnothing$, by definition of the closure.

(3) $\Rightarrow$ (2): Suppose otherwise, $\exists\, \varnothing \neq U \subseteq X$ open such that $\forall V \subseteq U$ nonempty open, $A \cap V \neq \varnothing$. Pick $x \in U$. Then $\forall W \ni x$ open, $W \cap U \subseteq U$ is open, thus $W \cap U \cap A \neq \varnothing$. Hence $A \cap W \neq \varnothing$ $\implies$

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$x \in \overline{A}$ by definition of the closure. Thus $U \subseteq \overline{A}$. Then $U \subseteq \operatorname{Int}(\overline{A})$, since $\operatorname{Int}(\overline{A})$ is the largest open set contained in $\overline{A}$. □

Recall. For a topological space X and $A, K \subseteq X$, the following are equivalent:

(i) K is the closure of A, $K = \overline{A}$.
(ii) $K = \bigcap\{C \subseteq X : C \text{ closed},\ A \subseteq C\}$.
(iii) K is the smallest closed set containing A.
(iv) $K = \{x \in X : \forall \text{ open } U \ni x,\ U \cap A \neq \varnothing\}$.

And for $A, I \subseteq X$, the following are equivalent:

(i) I is the interior of A, denoted $I = \operatorname{Int}(A)$.
(ii) $I = \bigcup\{U \subseteq X : U \text{ open},\ U \subseteq A\}$.
(iii) I is the largest open set contained in A.
(iv) $I = \{x \in X : \exists \text{ open } U \ni x,\ U \subseteq A\}$. □

Example 8.3

(a) The $\frac{1}{3}$-Cantor set $C \subseteq [0,1]$ is nowhere dense in $[0,1]$, because it is closed and has empty interior. (The size of C is $2^{\aleph_0}$!)
(b) Every compact subset $C \subseteq \mathbb{N}^{\mathbb{N}}$ is n.d.; in fact, it is closed and $\operatorname{Int}(C) = \varnothing$: C is closed $\iff$ $C = [T_C]$ (Proposition 3.7). By HW 1 (Proposition 3.11), $[T_C]$ is compact $\iff$ T_C is finitely branching, so C cannot contain a cylinder set in $\mathbb{N}^{\mathbb{N}}$.
(c) In a perfect Hausdorff space, every singleton is nowhere dense, because it is closed and has empty interior. □

Proposition 8.4

Let X be a topological space, $A, U \subseteq X$.

(a) A is n.d. $\iff$ $\overline{A}$ is n.d.

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(b) If U is open, then the boundary $\partial U := \overline{U} \setminus U$ is n.d.
(c) U is dense and open $\iff$ U^c is closed and n.d.
(d) N.d. sets form an ideal, i.e. they are closed under downward inclusion and under finite unions.

Proof. (a) is by Proposition 8.2, (1) $\iff$ (3).

(b) Since U is open, $\partial U = \overline{U} \setminus U = \overline{U} \cap U^c$ is closed. But ∂U has empty interior, by noticing that

$$\begin{aligned}\partial U = \overline{U} \setminus U &= \{x \in U^c : \forall V \ni x \text{ open},\ V \cap U \neq \varnothing\} \\ &= \{x \in X : \forall V \ni x \text{ open},\ V \cap U \neq \varnothing \text{ and } V \cap U^c \neq \varnothing\}.\end{aligned}$$

(c) U is dense open $\iff$ $\overline{U} = X$, U open $\iff$ $(\overline{U})^c = \varnothing$, U open $\iff$ $\operatorname{Int}(U^c) = \varnothing$, U^c closed $\iff$ U^c closed, U^c n.d.

(d) Let $A, B \subseteq X$ be n.d.; we show that $A \cup B$ is n.d. Fix open $U \neq \varnothing$. Then $\exists\, \varnothing \neq V \subseteq U$ such that $A \cap V = \varnothing$. But then $\exists\, \varnothing \neq W \subseteq V$ such that $B \cap W = \varnothing$, so $(A \cup B) \cap W = (A \cap W) \cup (B \cap W) = \varnothing \cup \varnothing = \varnothing$. Closure under downward inclusion is trivial by applying (3) of the previous proposition. Finite unions follow by induction. □

8.2 Meagre Sets

Completion of n.d. sets under countable unions.

Definition 8.5

In a topological space X, countable unions of n.d. sets are called meagre sets. Call a set comeagre if its complement is meagre. □

Proposition 8.6

Meagre subsets of X form a σ-ideal, i.e. they are closed under downward inclusion (i.e. subsets) and under countable

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unions. Equivalently, comeagre sets form a σ-filter, i.e. they are closed under supersets and under countable intersections.

Meagre sets are to topological spaces as null sets are to measure spaces.

Proposition 8.7

Let X be a topological space and $A \subseteq X$.

(a) A is meagre $\iff$ A is contained in a countable union of closed n.d. sets. In particular, every meagre set is contained in an F_σ meagre set.
(b) A is comeagre $\iff$ A contains a countable intersection of dense open sets. In particular, dense G_δ sets are comeagre.

Proof. (a) ($\Rightarrow$) A is meagre $\implies$ $A = \bigcup_{n\in\mathbb{N}} D_n$, each D_n n.d. $\implies$ $A \subseteq \bigcup_{n\in\mathbb{N}} \overline{D_n}$, each $\overline{D_n}$ closed n.d. ($\Leftarrow$) $A \subseteq \bigcup_{n\in\mathbb{N}} C_n$ $\implies$ $A = \bigcup_{n\in\mathbb{N}} (A \cap C_n)$, each $A \cap C_n$ n.d. by closure under subsets.

(b) follows from (a) and Proposition 8.4(c). □

Caution. The converse of (b), "every comeagre set contains a dense G_δ set", is not true. In fact, the whole space could be meagre, with two consequences: the complement $= \varnothing$ is comeagre, hence does not contain a dense G_δ set; and a set can be both meagre and comeagre. For instance, $\mathbb{Q}$ in its relative topology is meagre, because it is a countable union of singletons (closed n.d.)

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sets. In this case, a countable intersection of dense open sets could be empty (certainly not dense). E.g., enumerate $\mathbb{Q} = \{q_n\}_{n\in\mathbb{N}}$ and let $U_n := \mathbb{Q} \setminus \{q_i\}_{i=0}^n$; then each U_n is dense open in $\mathbb{Q}$, but $\bigcap_{n\in\mathbb{N}} U_n = \varnothing$.

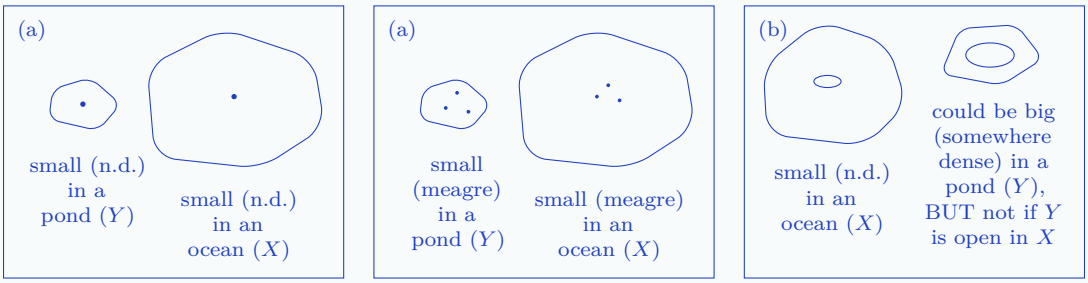
We will see that if we restrict to spaces with better properties (i.e. Baire spaces), then, as consequences of the Baire category theorem (Theorem 8.14), no set can be both meagre and comeagre, and every comeagre set must contain a dense G_δ set. □

8.3 Relativization of Nowhere Dense and Meagre

Proposition 8.8

Let X be a topological space, $Y \subseteq X$. Fix $A \subseteq Y$.

(a) If A is n.d. (respectively meagre) in Y, then it is still n.d. (respectively meagre) in X.
(b) If Y is open, then the converse holds as well: if A is n.d. (respectively meagre) in X, then A is n.d. (respectively meagre) in Y.



Proof. (a) It suffices to prove it for n.d. sets; then meagreness follows by taking countable unions. If A is n.d. in Y, then for every $U \subseteq X$ open with $U \cap Y \neq \varnothing$, there is $V \subseteq U$ open in X with $V \cap Y \neq \varnothing$ and $A \cap (V \cap Y) = \varnothing$. Then $A \cap (V \cap Y) = (A \cap Y) \cap V = A \cap V = \varnothing$ (and if $U \cap Y = \varnothing$, then U itself misses A). Thus by the equivalent conditions of n.d., A is n.d. in X.

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(b) Let $U \subseteq Y$ be nonempty open. Then $\exists \widetilde{U} \subseteq X$ open such that $\widetilde{U} \cap Y = U$. Since Y is open in X, so is U. Since A is n.d. in X, $\exists V \subseteq U$ open in X such that $V \cap A = \varnothing$. But $V = V \cap Y$ is open in Y with the relative topology. Thus by the equivalent condition, A is n.d. in Y. □

8.4 Baire Spaces ($\neq$ the Baire Space $\mathbb{N}^\mathbb{N}$)

We now give the class of topological spaces such that the σ-ideal $\mathrm{MGR}(X) := \{\text{meagre sets in } X\}$ doesn't trivialize, i.e. $\mathrm{MGR}(X) \subsetneq 2^X$, i.e. X is nonmeagre.

Definition 8.9

Call a topological space Baire if every nonempty open set in X is nonmeagre. □

Proposition 8.10

Let X be a topological space. The following are equivalent:

(1) X is Baire.
(2) Every comeagre set is dense in X.
(3) A countable intersection of dense open sets is a dense G_δ.

Proof. (1) $\Rightarrow$ (2): Let K be comeagre; then K^c is meagre. Let $U \neq \varnothing$ be open. Since X is Baire, $U \not\subseteq K^c$, since meagreness is closed under subsets; hence $U \cap K \neq \varnothing$. Thus K is dense.

(2) $\Rightarrow$ (3): Countable intersections of dense open sets are comeagre (Proposition 8.7(b)), hence dense.

(3) $\Rightarrow$ (2): Every comeagre subset contains a countable intersection of dense open sets (Proposition 8.7(b)), hence a dense G_δ. A set containing a dense subset is dense itself.

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(2) $\Rightarrow$ (1): Let $U \neq \varnothing$ be open. If U were meagre, then U^c is comeagre $\implies$ U^c is dense, impossible since U^c misses U. Hence X is Baire. □

Proposition 8.11

Given a Baire space X, the following hold:

(a) Given $X \neq \varnothing$, comeagre sets are nonmeagre.
(b) A set is comeagre $\iff$ it contains a dense G_δ set.
(c) Every open set is Baire with respect to the subspace topology.

Proof. (a) If U is meagre and comeagre, then $X = U \cup U^c$ is meagre. Contradiction.

(b) Every comeagre set contains a countable intersection of dense open sets, which is dense by Proposition 8.10(3). Conversely, every dense G_δ set is comeagre (Proposition 8.7(b)); comeagreness is closed under taking supersets, hence the set itself is comeagre.

(c) We prove the contrapositive. Suppose $\varnothing \neq U \subseteq X$ is not Baire with respect to the subspace topology; then there is a nonempty V that is open and meagre in U with respect to the relative topology. Then V is open in X since U is open, and meagre by relativization. Thus X is not Baire. □

Remark 8.12

The converse of (a) is not true: nonmeagre sets need not be comeagre; in other words, a set can be neither meagre nor comeagre. We will see that $\mathbb{R}$ is Baire by the Baire category theorem

(Theorem 8.14), since it is completely metrizable. $[0,1] \subseteq \mathbb{R}$ is nonmeagre: if $[0,1]$ is meagre, then

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$(0,1)$ is meagre. But $(0,1)$ is nonempty open in a Baire space, hence nonmeagre. Hence $[0,1]$ is nonmeagre. On the other hand, $[0,1]^c$ is open, thus nonmeagre. Thus $[0,1]$ is neither meagre nor comeagre in $\mathbb{R}$. □

Corollary 8.13

In Baire spaces, a set is comeagre $\iff$ it contains a dense G_δ set.

Proof. Recall that a set is comeagre $\iff$ it contains a countable intersection of dense open sets (Proposition 8.7(b)), which is a dense G_δ in Baire spaces (Proposition 8.10(3)). □

8.5 Baire Category Theorem

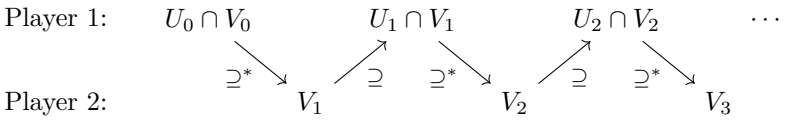
Theorem 8.14 (Baire category theorem)

Completely metrizable topological spaces are Baire. In particular, Polish spaces are Baire. Locally compact Hausdorff spaces are Baire.

Proof. Complete metric spaces. Let $(U_n)_{n\in\mathbb{N}}$ be a sequence of dense open sets and $V_0 \subseteq X$ a nonempty open set.

We want to show: $\bigcap_{n\in\mathbb{N}} U_n \cap V_0 \neq \varnothing$, i.e. every countable intersection of dense open sets is dense (by Proposition 8.10 characterizing Baire spaces, (1) $\iff$ (3)).

We will play a 2-player game and apply the nested closed sets property of a completely metrizable space:



Recall (notation from §7) $A \subseteq^* B :\iff \overline{A} \subseteq B$, and we further ask each V_n to be nonempty open with $\operatorname{diam}(V_n) \leq \frac{1}{n}$, $n \geq 1$.

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Granted such a game,

$$\bigcap_{n\in\mathbb{N}} U_n \cap V_0 \supseteq \bigcap_{n\geq 1} \overline{V_n},$$

where the $\overline{V_n}$ are all nonempty and $\operatorname{diam}(\overline{V_n}) = \operatorname{diam}(V_n) \xrightarrow{n\to\infty} 0$. Thus by HW 1, Q2, $\bigcap_{n\in\mathbb{N}} U_n \cap V_0 \supseteq \bigcap_{n\geq 1} \overline{V_n} \neq \varnothing$. Hence it follows that X is Baire.

It remains to show that the two players always have a feasible move. Given a move from player 2, player 1 simply takes the intersection, which is nonempty since each U_n is dense in X. Given a move from

player 1, player 2 also has a move: given open $U_i \cap V_i \neq \varnothing$, $\exists\, B_\varepsilon(x) \subseteq U_i \cap V_i$; then take $B_{\varepsilon/2}(x) =: V_{i+1}$. We can verify

$$\overline{V_{i+1}} = \overline{B_{\varepsilon/2}(x)} \subseteq B_{\varepsilon/2}[x] \subseteq B_\varepsilon(x) \subseteq U_i \cap V_i,$$

where $\overline{B_{\varepsilon/2}(x)} \subseteq B_{\varepsilon/2}[x]$ since $B_{\varepsilon/2}[x]$ is closed, $B_{\varepsilon/2}(x) \subseteq B_{\varepsilon/2}[x]$, and $\overline{B_{\varepsilon/2}(x)}$ is the smallest closed set containing $B_{\varepsilon/2}(x)$.

LCH spaces. Run the same 2-player game, except instead of requiring $\operatorname{diam}(V_n) \leq \frac{1}{n}$, we let player 2 pick $\varnothing \neq V_{n+1} \subseteq^* U_n \cap V_n$ with $\overline{V_{n+1}}$ compact (feasible by the shrinking lemma, Lemma 7.3). Then $\bigcap_{n\in\mathbb{N}} U_n \cap V_n \supseteq \bigcap_{n\in\mathbb{N}} \overline{V_{n+1}}$; apply the following lemma (Lemma 8.15), and Baireness follows from nonemptiness. □

Lemma 8.15

If $\{K_n\}_{n=1}^\infty$ is a decreasing collection of compact nonempty sets in a Hausdorff space, then $\bigcap_{n\in\mathbb{N}^+} K_n \neq \varnothing$.

Proof. Notice that $\{K_n\}_n$ is a family of closed (this is the only place where we need Hausdorffness) subsets of the compact set K_1. Furthermore, it satisfies the finite intersection

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property: any finite intersection of K_n's = the smallest K_n among the finite collection $\neq \varnothing$. Thus by the equivalent definition of compactness (via the finite intersection property, Definition 4.1), K_1 compact $\implies \bigcap_{n\in\mathbb{N}^+} K_n \neq \varnothing$. □

Corollary 8.16

In nonempty Polish spaces, dense meagre sets are not G_δ. In particular, $\mathbb{Q}$ is not a G_δ subset of $\mathbb{R}$.

Proof. Dense G_δ sets are comeagre (always, Proposition 8.7(b)), hence nonmeagre (Proposition 8.11(a)) since Polish spaces are Baire (Theorem 8.14). In particular, $\mathbb{R}$ is Polish, and $\mathbb{Q}$ is dense meagre since $\mathbb{Q}$ is itself a countable union of closed n.d. sets (singletons). □

Definition 8.17

For a Baire space X and $P \subseteq X$ (a property), we say that P holds generically, or a generic point satisfies P, if P is comeagre. □

Note 8.18 (Idea)

Like with cardinality / probability / measure, comeagreness gives a method of proving the existence of points with a desired property. Namely, we prove the stronger statement that such points are generic, i.e. form a comeagre set.

This is similar to Cantor's proof of the existence of transcendental numbers: every algebraic number is a root of some $p(x) \in \mathbb{Q}[x]$, and $\mathbb{Q}[x]$ is countable. $\mathbb{R}$ is uncountable. Thus the collection of transcendental numbers must be uncountable. In contrast, Liouville constructed such a class of

transcendental numbers, called the Liouville numbers, with the so-called

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Liouville property, and the first explicit transcendental number $\sum_{n=1}^{\infty} 10^{-n!}$. Indeed, the set L of all Liouville numbers has size continuum $\mathfrak{c}$ and measure 0, and he proved that the algebraic numbers cannot satisfy the Liouville property. Yet the construction is expensive if we only need to show existence. □

9 Summary: Universal and Canonical Spaces

Part 1 keeps returning to three canonical spaces, the Cantor space $2^{\mathbb{N}}$, the Baire space $\mathbb{N}^{\mathbb{N}}$ and the Hilbert cube $[0,1]^{\mathbb{N}}$ (with $\mathbb{R}^{\mathbb{N}}$ alongside it), and to a handful of theorems describing how an arbitrary Polish space relates to them. This section collects those results in one place. The last column of each table points to the result in these notes where the statement is proved; the numbers are clickable links.

Big spaces: universal for embeddings.

Space	Statement	Where
$[0,1]^{\mathbb{N}}$ Hilbert cube	Every separable metrizable space embeds into $[0,1]^{\mathbb{N}}$. Hence, up to homeomorphism, Polish spaces = the G_δ subsets of $[0,1]^{\mathbb{N}}$, and compact metrizable spaces = the closed subsets.	Theorem 4.9
$\mathbb{R}^{\mathbb{N}}$	Every Polish space is homeomorphic to a closed subset of $\mathbb{R}^{\mathbb{N}}$, so $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$ "=" the set of all Polish spaces.	Theorem 4.11
$\mathbb{N}^{\mathbb{N}}$ zero-dimensional only	Every zero-dimensional separable metrizable space embeds into $\mathbb{N}^{\mathbb{N}}$; every zero-dimensional Polish space is homeomorphic to a closed subset of $\mathbb{N}^{\mathbb{N}}$, i.e. to some $[T]$ with $T \subseteq \mathbb{N}^{<\mathbb{N}}$ a pruned tree.	Theorem 6.15
$2^{\mathbb{N}}$ coding	Every Polish space (second countable Hausdorff suffices) admits an injection $b : X \hookrightarrow 2^{\mathbb{N}}$ with $b^{-1}([n \mapsto 1])$ open for every n: a Borel injection, not a topological embedding.	Proposition 2.6

Related. Alexandrov's theorem (Theorem 2.9) says which subspaces of a Polish space are Polish: exactly the G_δ ones. That is what turns "embeds into the cube" into "= a G_δ subset of the cube".

Small spaces: universal for surjections.

Space	Statement	Where
$2^{\mathbb{N}}$ Cantor space	Every nonempty compact metrizable space is a continuous image of $2^{\mathbb{N}}$. (Route: $2^{\mathbb{N}} \twoheadrightarrow [0,1]$ by binary expansion, then $(2^{\mathbb{N}})^{\mathbb{N}} \cong 2^{\mathbb{N}} \twoheadrightarrow [0,1]^{\mathbb{N}}$, then a retraction of $2^{\mathbb{N}}$ onto the closed preimage of the given space.)	Theorem 4.12
$\mathbb{N}^{\mathbb{N}}$ Baire space	Every nonempty Polish space is a continuous image of $\mathbb{N}^{\mathbb{N}}$; in fact it is a continuous *bijective* image of some closed $C \subseteq \mathbb{N}^{\mathbb{N}}$.	Theorem 6.14

So $2^{\mathbb{N}}$ is in general not homeomorphic to a given compact metrizable space (it is zero-dimensional, while $[0,1]$ is not): it only maps *onto* it.

Small spaces inside everything of a kind: dichotomies.

Statement	Where
Perfect set theorem. Every nonempty perfect Polish space contains a homeomorphic copy of $2^{\mathbb{N}}$ (built by a Cantor scheme).	Theorem 5.4
Cantor–Bendixson. Every Polish space = (countable open set) $\sqcup$ (perfect kernel), obtained from the iterated derivatives X^{α}, which stabilize at a countable ordinal.	Theorems 5.6, 5.12
Perfect set property. Every Polish space is countable or contains a copy of $2^{\mathbb{N}}$; hence its cardinality is $\leq \aleph_0$ or $2^{\aleph_0}$ (the continuum hypothesis holds for Polish spaces). Extended to Borel sets in Part 2.	Theorem 5.5
Hurewicz dichotomy. Every Polish space is σ-compact or contains a closed copy of $\mathbb{N}^{\mathbb{N}}$.	Theorem 3.13

Characterizations: the canonical spaces are unique.

Statement	Where
Brouwer. $2^{\mathbb{N}}$ is the unique nonempty perfect compact zero-dimensional metrizable space (so the Cantor set is $\cong 2^{\mathbb{N}}$).	Theorem 6.11
Alexandrov–Urysohn. $\mathbb{N}^{\mathbb{N}}$ is the unique nonempty zero-dimensional Polish space in which every compact set has empty interior (so $\mathbb{R} \setminus \mathbb{Q} \cong \mathbb{N}^{\mathbb{N}}$).	Theorem 6.12
Closed sets are trees. The closed subsets of $A^{\mathbb{N}}$ are exactly the $[T]$ for pruned trees $T \subseteq A^{<\mathbb{N}}$, and $[T]$ is compact iff T is finitely branching (Kőnig).	Propositions 3.7, 3.11

How the three canonical spaces relate to each other.

- $2^{\mathbb{N}} \subseteq \mathbb{N}^{\mathbb{N}}$ is a closed (indeed compact) subset; conversely, $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to a G_δ subset of $2^{\mathbb{N}}$ (Example 2.12, Theorem 6.15).
- $2^{\mathbb{N}} \hookrightarrow [0,1]$ with image the Cantor set, and $2^{\mathbb{N}} \twoheadrightarrow [0,1]$ by binary expansion (Example 4.8).
- $\mathbb{N}^{\mathbb{N}} \cong \mathbb{R} \setminus \mathbb{Q}$ (Corollary 6.13).
- $(2^{\mathbb{N}})^{\mathbb{N}} \cong 2^{\mathbb{N}}$ (proof of Theorem 4.12), and likewise $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \cong \mathbb{N}^{\mathbb{N}}$. Both are used constantly in Part 2 to encode countably many codes as a single one.

Note 9.1 (The mental picture)

$[0,1]^{\mathbb{N}}$ and $\mathbb{R}^{\mathbb{N}}$ are the big ambient spaces everything lives in. $\mathbb{N}^{\mathbb{N}}$ is the universal parametrizer: it maps onto every Polish space and contains every zero-dimensional one as a closed subset. $2^{\mathbb{N}}$ is its compact cousin: it maps onto every compact metrizable space and embeds into every uncountable Polish space. Conversely, every Polish space injects into $2^{\mathbb{N}}$ by binary representation (Proposition 2.6), but in general only as a Borel injection, not as an embedding. In the picture, blue hooked arrows $\hookrightarrow$ are injections: solid for embeddings, dashed for the binary representation. Brown double-headed arrows $\twoheadrightarrow$ are continuous surjections, and the ovals are the classes of spaces, with a few members written inside.

